

Technology – Electronic Components

Industry data

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See the end pages of this presentation for analyst certification and important disclosures, including non-US analyst disclosures.

End demand assumptions

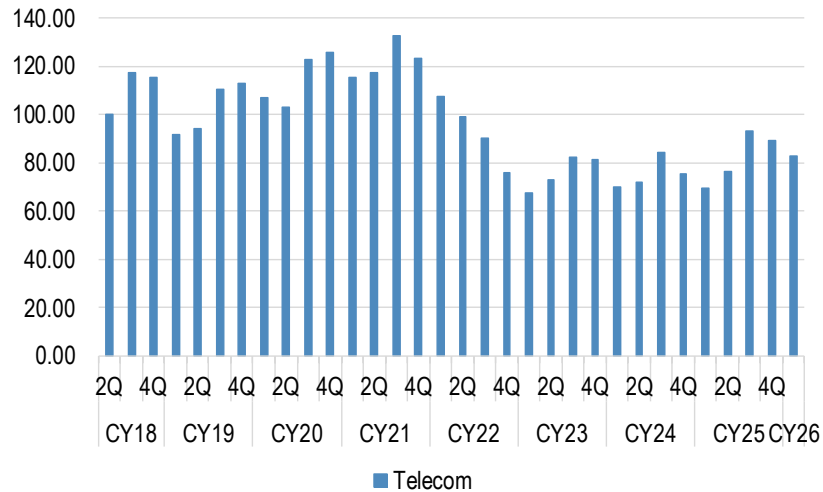
FY (mn units)	CY2020	CY2021	CY2022	CY2023	CY2024	CY2025	CY2026E	CY2027E	CY2025	1Q	2Q	3Q	4Q	CY2026	1QE	2QE	3QE	4QE
End demand volume (shipment)																		
Global smartphone shipment	1,264.74	1,351.40	1,193.43	1,141.86	1,223.18	1,245.45	1,104.26	1,075.13	296.94	288.92	320.06	339.53	292.13	262.70	272.45	276.97		
yoy % chg.	-	6.85%	-11.69%	-4.32%	7.12%	1.82%	-11.34%	-2.64%	0.24%	-0.01%	3.26%	3.49%	-1.62%	-9.07%	-14.87%	-18.43%		
qoq % chg.	-	-	-	-	-	-	-	-	-9.49%	-2.70%	10.78%	6.09%	-13.96%	-10.08%	3.71%	1.66%		
Apple	207.15	230.06	232.16	229.14	225.85	240.64	238.55	252.50	55.01	44.79	56.50	84.33	60.41	49.69	56.79	71.66		
yoy % chg.	-	11.06%	0.91%	-1.30%	-1.44%	6.55%	-0.87%	5.85%	12.95%	-1.77%	3.76%	9.39%	9.81%	10.93%	0.51%	-15.02%		
qoq % chg.	-	-	-	-	-	-	-	-	-28.64%	-18.58%	26.15%	49.24%	-28.36%	-17.75%	14.29%	26.18%		
Samsung	255.52	274.50	257.95	225.46	222.91	239.13	230.93	228.50	60.55	57.50	60.64	60.45	59.00	57.37	59.10	55.46		
yoy % chg.	-	7.43%	-6.03%	-12.59%	-1.13%	7.28%	-3.43%	-1.05%	0.95%	7.40%	5.54%	16.38%	-2.55%	-0.22%	-2.55%	-8.25%		
qoq % chg.	-	-	-	-	-	-	-	-	16.57%	-5.03%	5.46%	-0.32%	-2.39%	-2.76%	3.00%	-6.15%		
Others	802.08	846.73	703.31	687.27	774.44	765.62	634.79	594.08	181.54	186.83	202.62	194.17	172.91	155.70	156.42	149.74		
yoy % chg.	-	5.57%	-16.94%	-2.28%	12.68%	-1.14%	-17.09%	-6.41%	-3.48%	-1.56%	2.46%	-2.25%	-4.75%	-16.66%	-22.80%	-22.88%		
qoq % chg.	-	-	-	-	-	-	-	-	-8.61%	2.92%	8.45%	-4.17%	-10.95%	-9.95%	0.46%	-4.27%		
Global PC shipment	308.17	341.73	284.20	242.28	247.65	268.63	245.71	249.53	60.32	65.94	72.46	69.91	65.68	59.50	60.00	60.53		
yoy % chg.	-	10.89%	-16.83%	-14.75%	2.21%	8.47%	-8.53%	1.55%	6.58%	8.05%	12.15%	6.88%	8.89%	-9.77%	-17.19%	-13.42%		
qoq % chg.	-	-	-	-	-	-	-	-	-7.78%	9.33%	9.88%	-3.52%	-6.05%	-9.41%	0.85%	0.88%		
Global Notebook PC (Including Chrome)	228.33	256.38	208.12	179.83	186.54	201.52	183.79	186.35	44.96	49.60	54.64	52.32	49.16	44.39	44.92	45.32		
yoy % chg.	-	12.29%	-18.82%	-13.59%	3.73%	8.03%	-8.80%	1.39%	7.14%	7.72%	11.19%	5.95%	9.33%	-10.50%	-17.78%	-13.39%		
qoq % chg.	-	-	-	-	-	-	-	-	-8.97%	10.32%	10.16%	-4.23%	-6.06%	-9.69%	1.19%	0.88%		
Global Desktop PC	79.84	85.34	76.08	62.45	61.11	67.11	61.93	63.18	15.36	16.34	17.82	17.59	16.52	15.11	15.08	15.21		
yoy % chg.	-	6.89%	-10.85%	-17.91%	-2.15%	9.82%	-7.72%	2.03%	4.96%	9.07%	15.19%	9.75%	7.59%	-7.55%	-15.38%	-13.49%		
qoq % chg.	-	-	-	-	-	-	-	-	-4.14%	6.40%	9.06%	-1.33%	-6.04%	-8.57%	-0.17%	0.87%		
Global Server shipment	12.67	12.92	13.83	11.35	12.09	12.12	13.88	15.03	2.94	2.94	2.86	3.38	3.26	3.40	3.44	3.78		
yoy % chg.	-	1.94%	7.04%	-17.90%	6.46%	0.25%	14.57%	8.27%	2.94%	-1.53%	-5.62%	5.05%	11.13%	15.63%	20.17%	11.88%		
qoq % chg.	-	-	-	-	-	-	-	-	-8.68%	0.15%	-2.68%	18.03%	-3.39%	4.20%	1.14%	9.88%		
General Server	12.67	12.92	13.43	10.78	11.02	10.30	11.68	12.44	-	-	-	-	-	-	-	-		
yoy % chg.	-	1.94%	3.94%	-19.75%	2.27%	-6.49%	13.39%	6.51%	-	-	-	-	-	-	-	-		
qoq % chg.	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-		
AI server	-	-	0.40	0.58	1.07	1.81	2.20	2.58	-	-	-	-	-	-	-	-		
yoy % chg.	-	-	-	44.08%	84.66%	69.87%	21.28%	17.63%	-	-	-	-	-	-	-	-		
qoq % chg.	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-		
IHS-Global Auto Production	74.59	77.19	82.34	90.49	89.59	93.10	90.88	91.58	22.26	22.92	23.10	24.82	21.86	22.44	22.70	23.88		
yoy % chg.	-	3.49%	6.66%	9.90%	-0.99%	3.92%	-2.38%	0.77%	2.94%	-1.53%	-5.62%	5.05%	11.13%	15.63%	20.17%	11.88%		
qoq % chg.	-	-	-	-	-	-	-	-	-8.68%	0.15%	-2.68%	18.03%	-3.39%	4.20%	1.14%	9.88%		
Global Vehicle Sales	77.18	80.27	78.98	86.70	88.69	91.74	87.69	88.55	-	-	-	-	-	-	-	-		
yoy % chg.	-	3.99%	-1.60%	9.76%	2.30%	3.44%	-4.42%	0.99%	-	-	-	-	-	-	-	-		
qoq % chg.	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-		
ICE	71.52	68.12	62.17	64.15	60.90	58.04	52.93	51.29	-	-	-	-	-	-	-	-		
yoy % chg.	-	-4.77%	-8.73%	3.20%	-5.07%	-4.70%	-8.80%	-3.10%	-	-	-	-	-	-	-	-		
qoq % chg.	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-		
xEV	5.66	12.15	16.82	22.54	27.79	33.70	34.76	37.26	-	-	-	-	-	-	-	-		
yoy % chg.	-	114.69%	38.42%	34.04%	23.27%	21.28%	3.13%	7.21%	-	-	-	-	-	-	-	-		
qoq % chg.	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-		
HEV	2.72	5.71	6.22	8.16	9.84	11.07	12.39	13.92	-	-	-	-	-	-	-	-		
yoy % chg.	-	110.01%	9.00%	31.17%	20.52%	12.57%	11.92%	12.33%	-	-	-	-	-	-	-	-		
qoq % chg.	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-		
PHEV	0.89	1.82	2.74	4.08	6.47	7.61	6.95	7.36	-	-	-	-	-	-	-	-		
yoy % chg.	-	104.61%	51.14%	48.70%	58.63%	17.48%	-8.62%	5.91%	-	-	-	-	-	-	-	-		
qoq % chg.	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-		
BEV	2.04	4.61	7.83	10.29	11.47	15.01	15.40	15.96	-	-	-	-	-	-	-	-		
yoy % chg.	-	125.40%	69.98%	31.32%	11.48%	30.89%	2.58%	3.66%	-	-	-	-	-	-	-	-		
qoq % chg.	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-		
FCV	0.01	0.02	0.02	0.01	0.01	0.01	0.01	0.02	-	-	-	-	-	-	-	-		
yoy % chg.	-	86.31%	-0.36%	-40.88%	-36.62%	41.82%	35.00%	40.00%	-	-	-	-	-	-	-	-		
qoq % chg.	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-		

Source: IDC (Smartphone), Gartner® (PC and Server), S&P Global (Automotive) and J.P. Morgan estimates

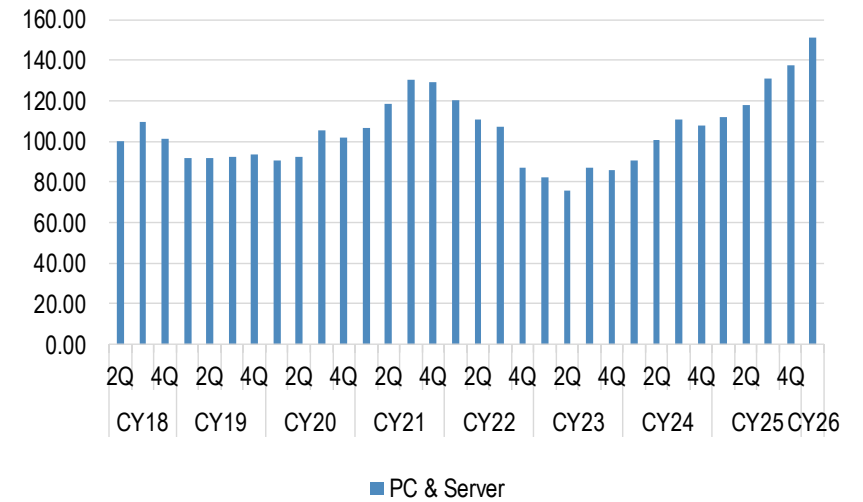
Electronic components

Sales index by application

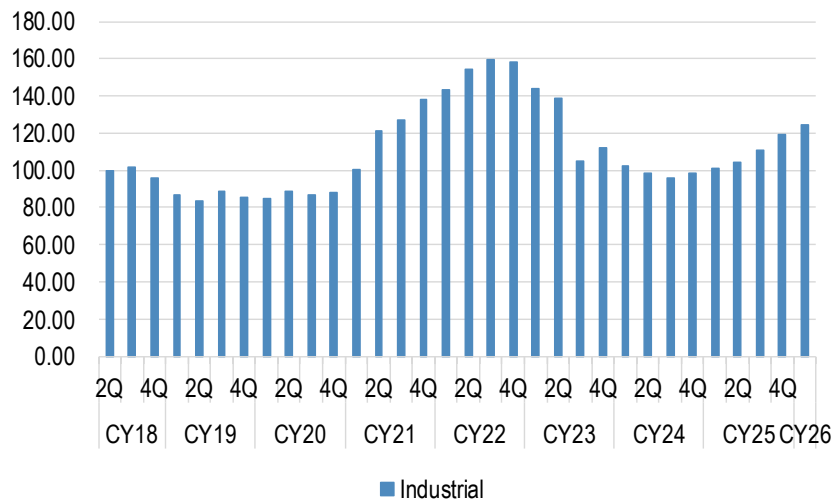
For Telecom (2Q18=100)



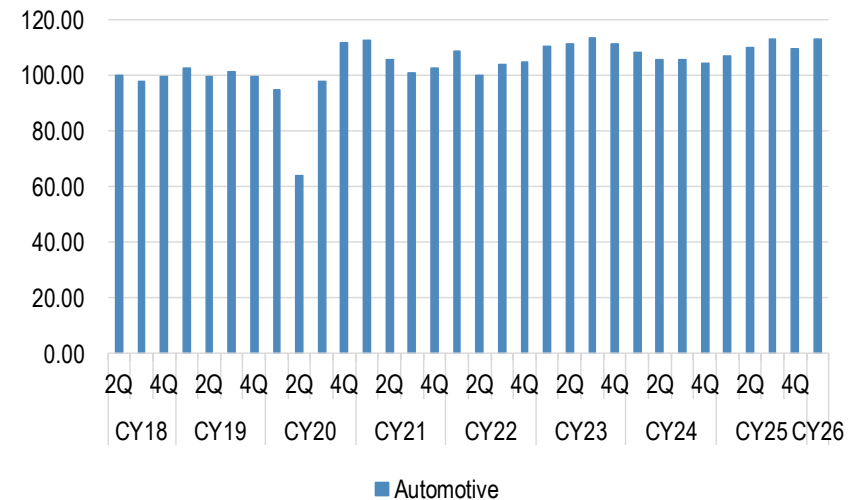
For PC & Servers (including FCBGA, HDD related, 2Q18=100)



For Industrial Applications (2Q18=100)



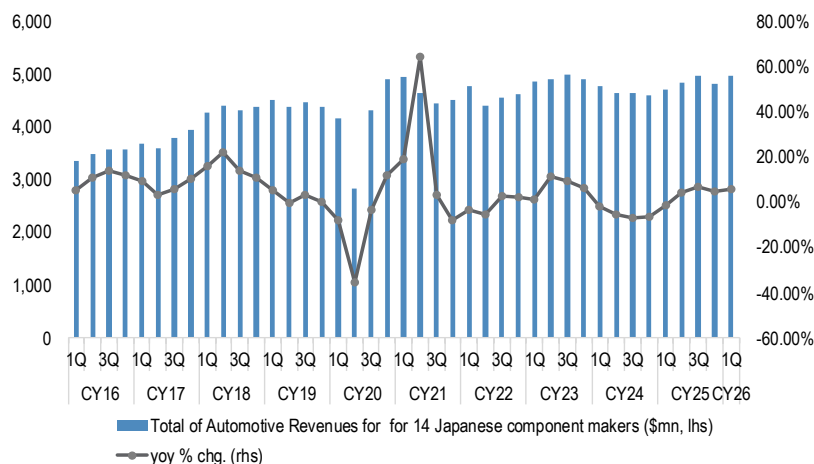
For Automotive (2Q18=100)



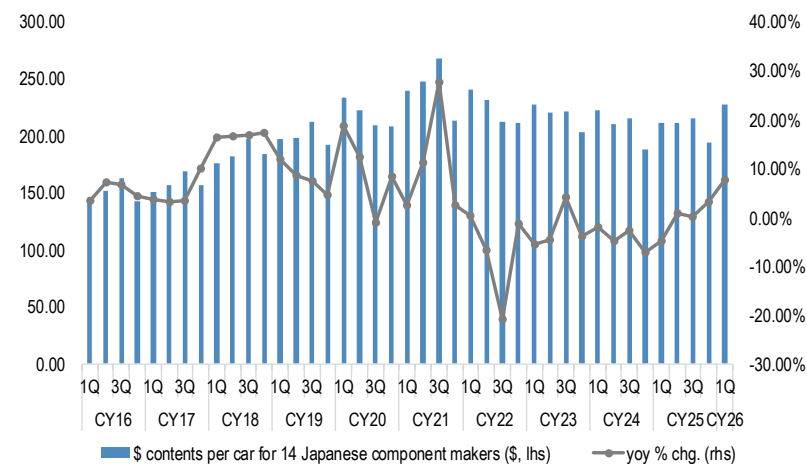
Source: Based on company data from Murata Mfg., TDK, Taiyo Yuden, Rohm, Nidec, MinebeaMitsumi, Hirose Electric, JAE, Ibiden, Kyocera, Nichicon,

Quarterly automotive component revenues and \$ contents per car, comparison with theoretical contents growth

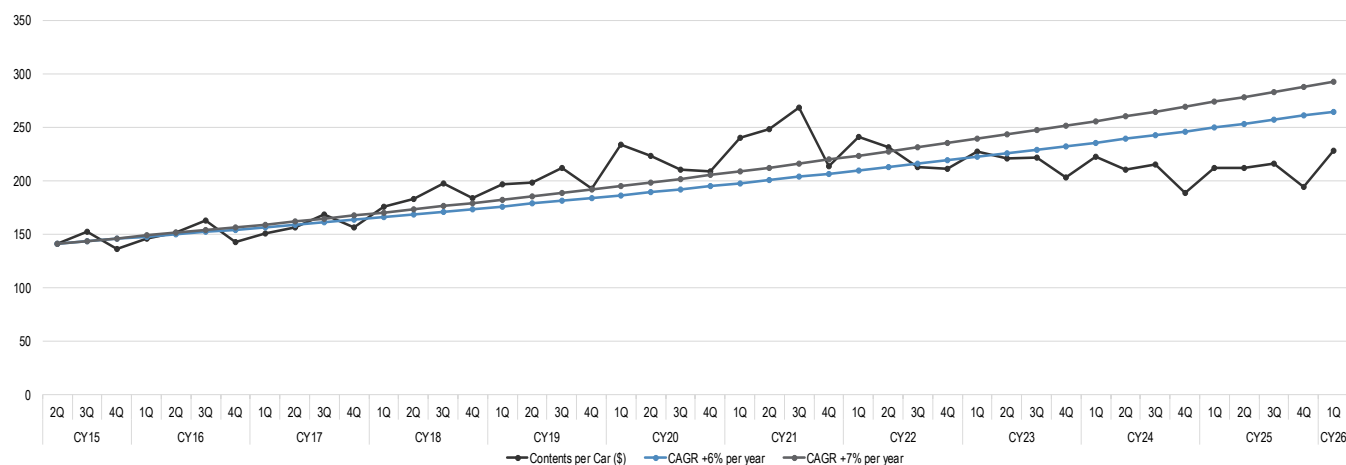
Automotive component revenues for 14 Japanese component makers



\$ contents per car for 14 Japanese component makers



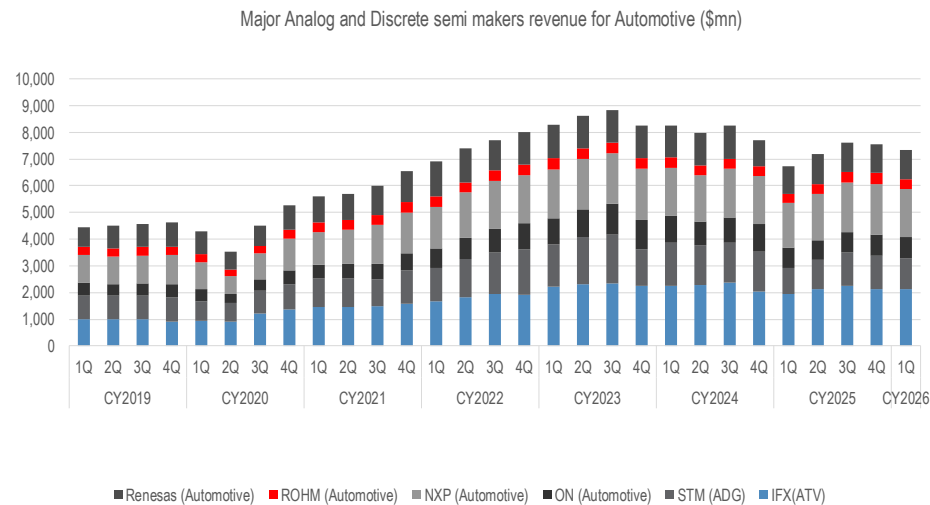
Comparison between \$ contents per car for 14 Japanese component makers and theoretical \$ contents trend



Source: Company data from Iriden, Niterra, Mabuchi Motor, Murata Mfg., TDK, Taiyo Yuden, Nichicon, Nippon Chemi-con, Nippon Denpa Kogyo, Alps Alpine, Hirose Electric, JAE, Yokowo, Rohm, J.P. Morgan

Comparison of automotive semiconductors and electronic components

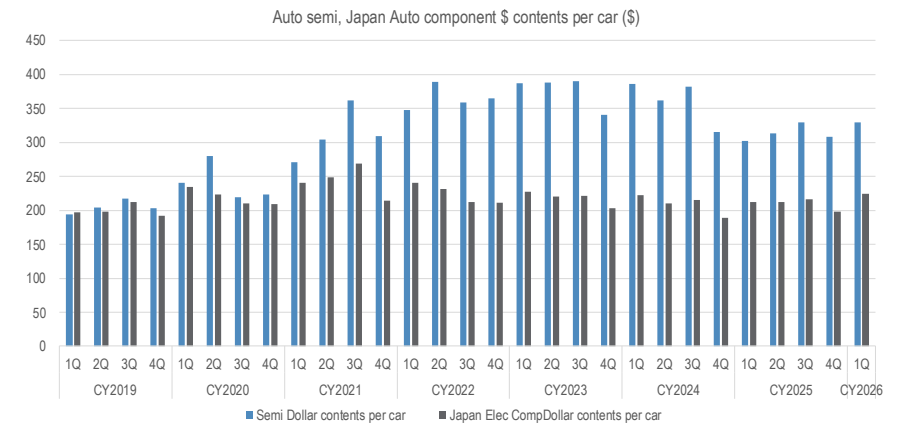
Automotive SEMI revenues for major analog/discrete/MCU makers



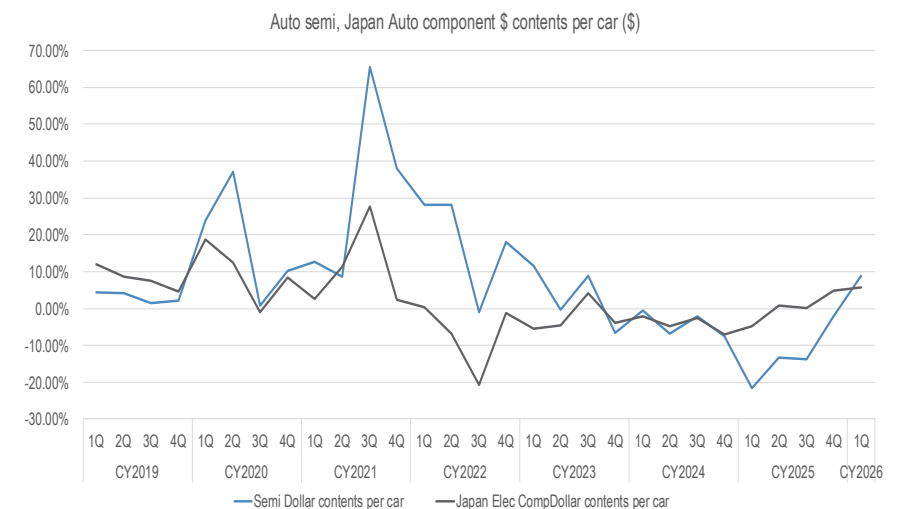
Note: For semiconductors, it is the total of IFX, STM, ON Semiconductor, NXP, Rohm, and Renesas Electronics; for Japanese electronic components, it is the total of Ibiden, Nittera, Mabuchi Motor, Murata Mfg., TDK, Taiyo Yuden, Nichicon, Nippon Chemi-con, Nippon Denpa Kogyo, Alps Alpine, Hirose Electric, JAE, and Yokowo.

Source: Company data, J.P. Morgan

\$ contents comparison for major analog/discrete/MCU makers and 14 Japanese component makers

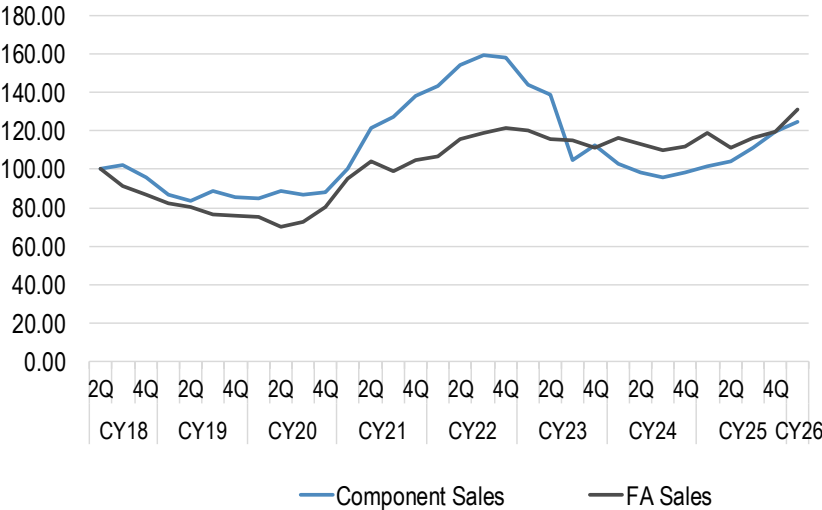


YoY % change of \$ contents for major analog/discrete/MCU makers and 14 Japanese component makers

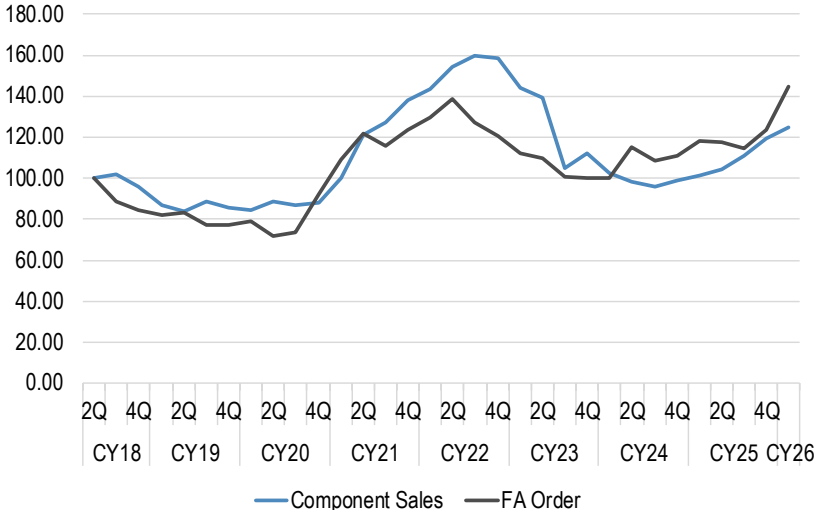


FA makers sales/orders vs. industrial component sales

FA makers sales and industrial component sales
index (2Q18=100)

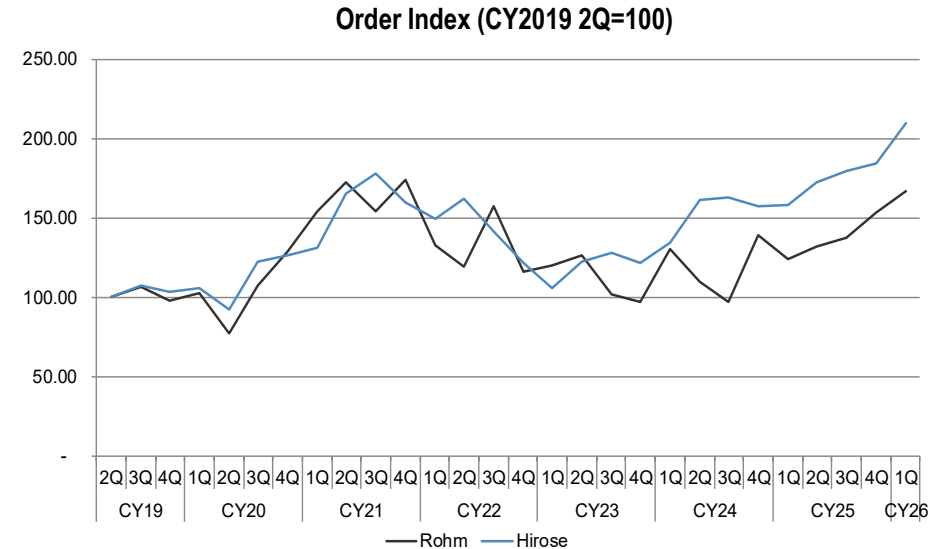
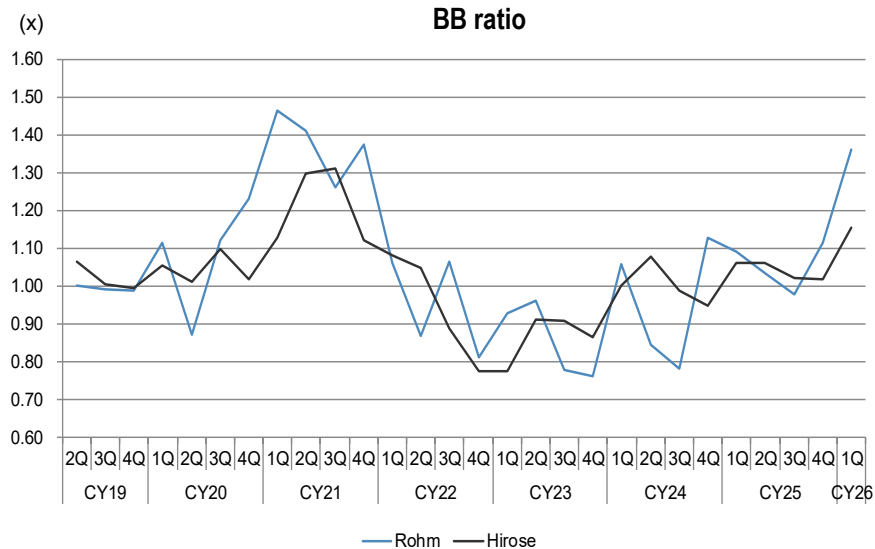
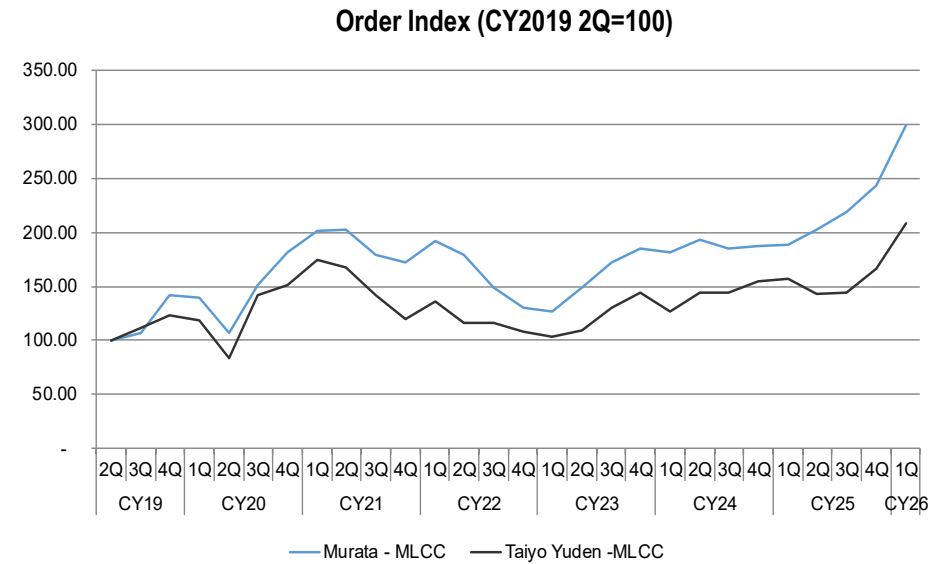
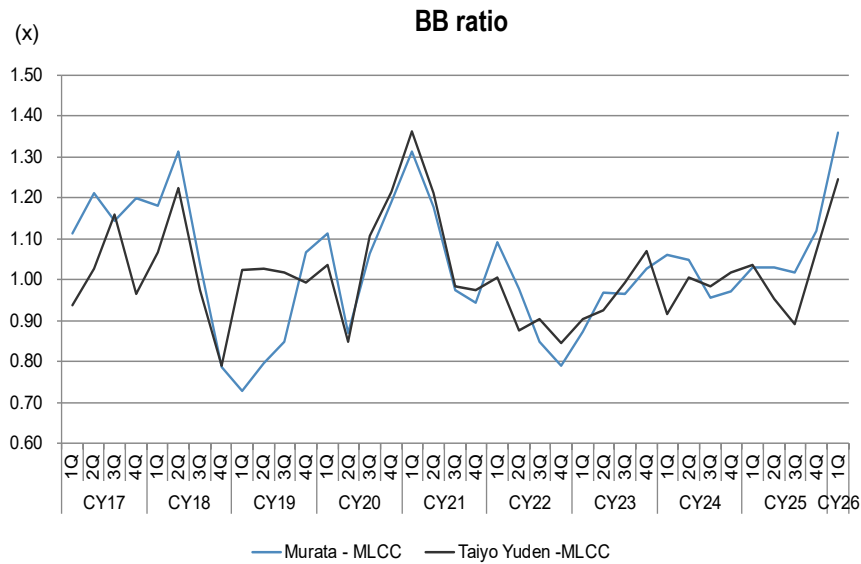


FA makers orders and industrial component sales
index (2Q18=100)



Note: FA manufacturers are Fanuc, SMC, and Yaskawa Electric; the electronic components are from Murata Manufacturing, TDK, Taiyo Yuden, Rohm, Nidec, MinebeaMitsumi, Hirose, JAE, Ibiden, Kyocera, Nichicon, Nippon Chemi-con, Niterra, and Nissha.

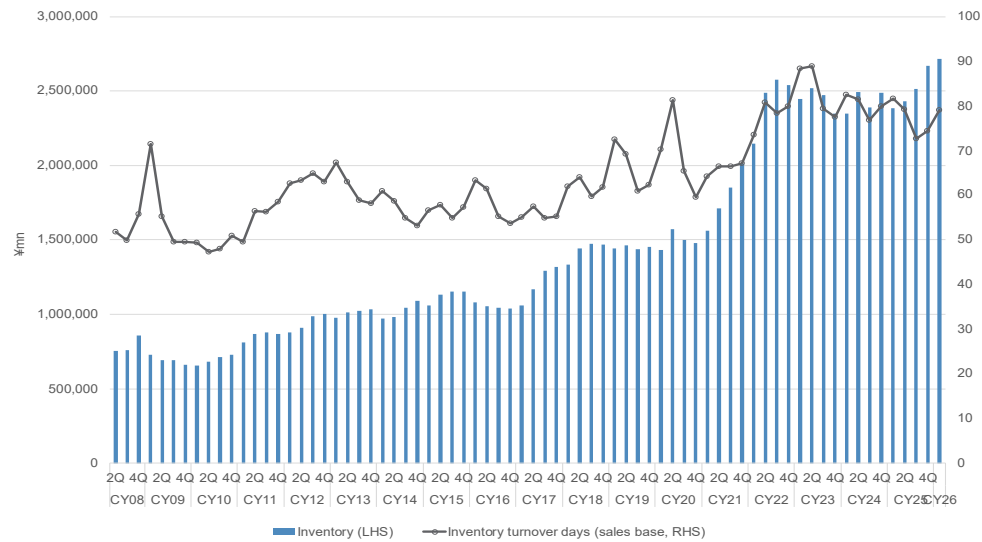
BB ratio and order index for Murata, Taiyo Yuden, Hirose, Rohm



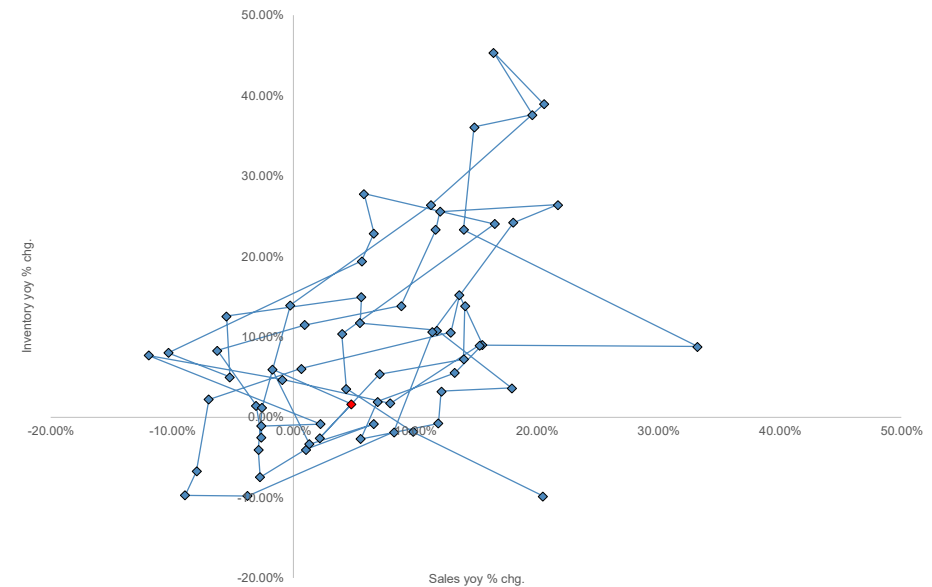
Source: Company data, J.P. Morgan.

Inventory and DIO for Japanese components sector

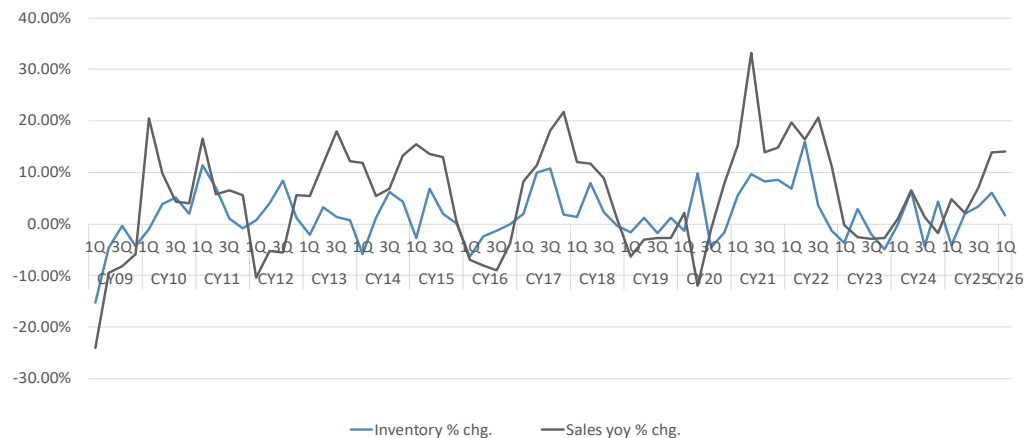
Inventories / DIO



Inventories yoy % chg.

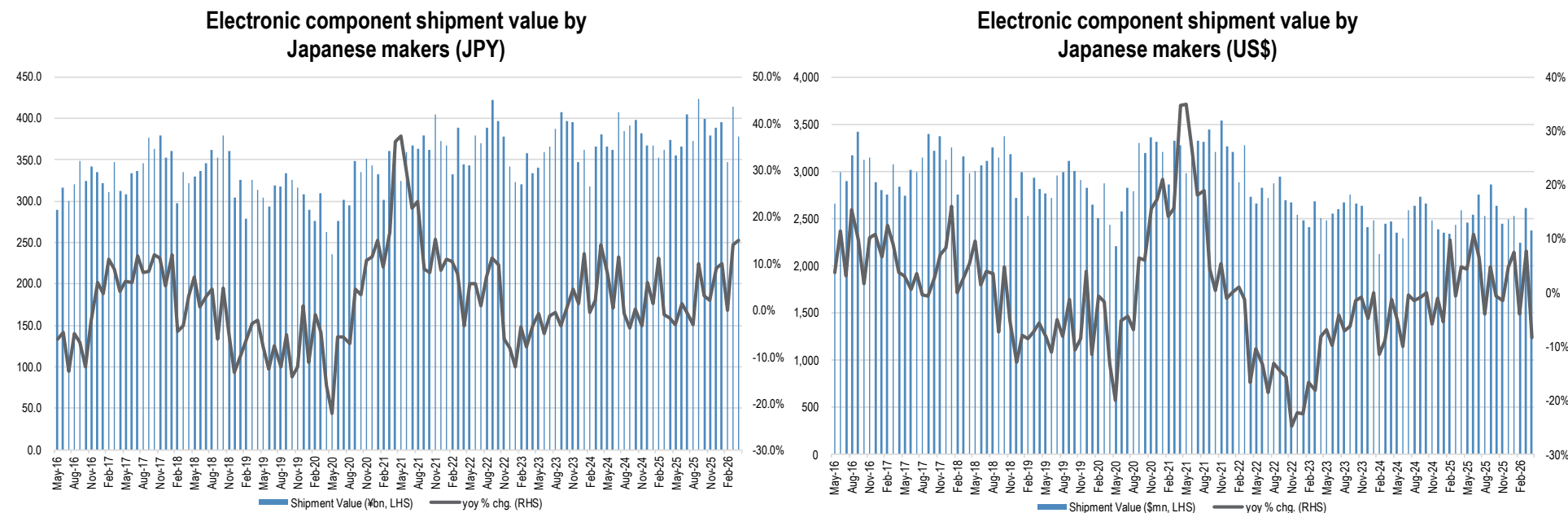


Inventories and sales yoy % chg.



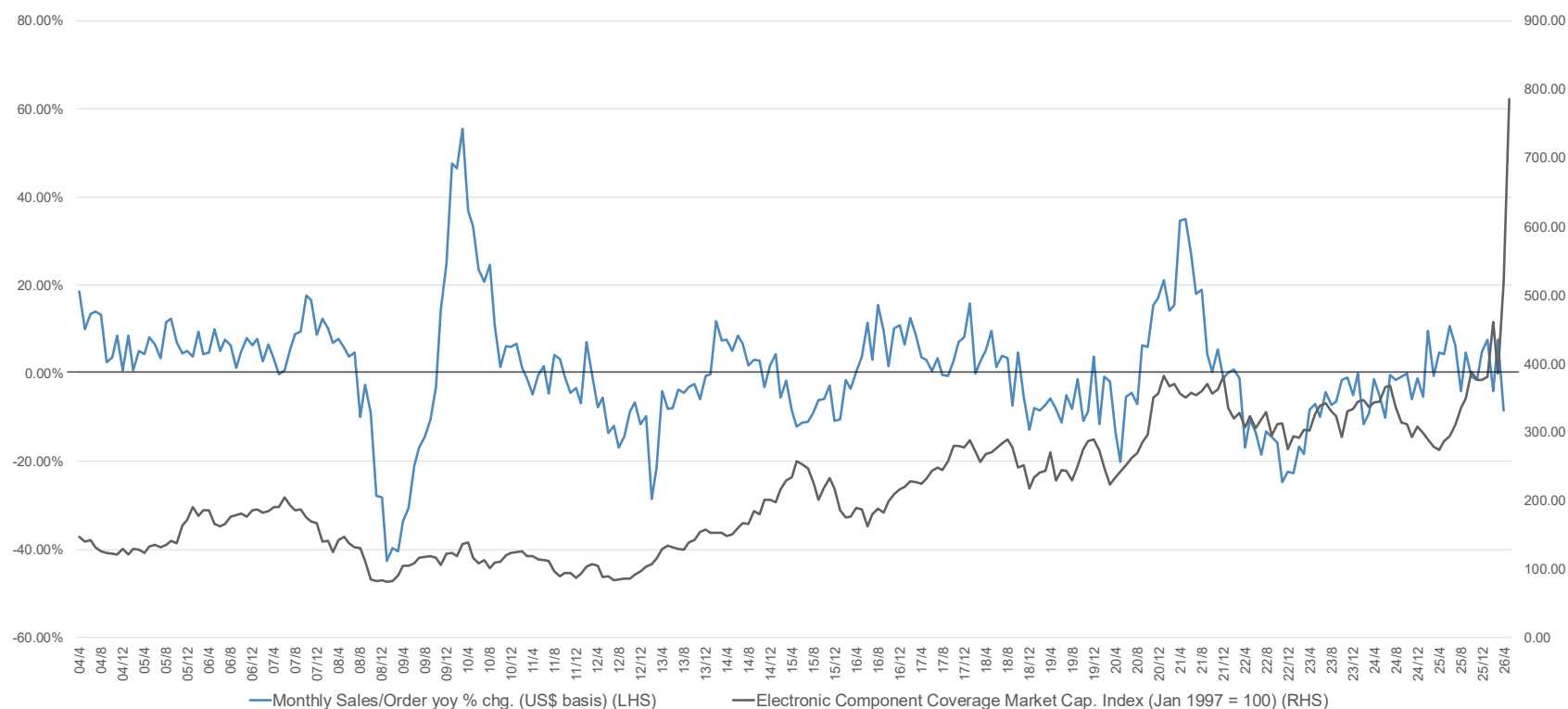
Note: Inventory and Sales include Murata Mfg., Taiyo Yuden, TDK, Kyocera, Rohm, Nichicon, Nippon Chemi-Con, MinebeaMitsumi, Ibiden, Hirose Electric, JAE, Alps Alpine, Niterra, Wacom, NISSHA.
Source: Bloomberg Finance L.P., JEITA, J.P. Morgan

JEITA global component shipments



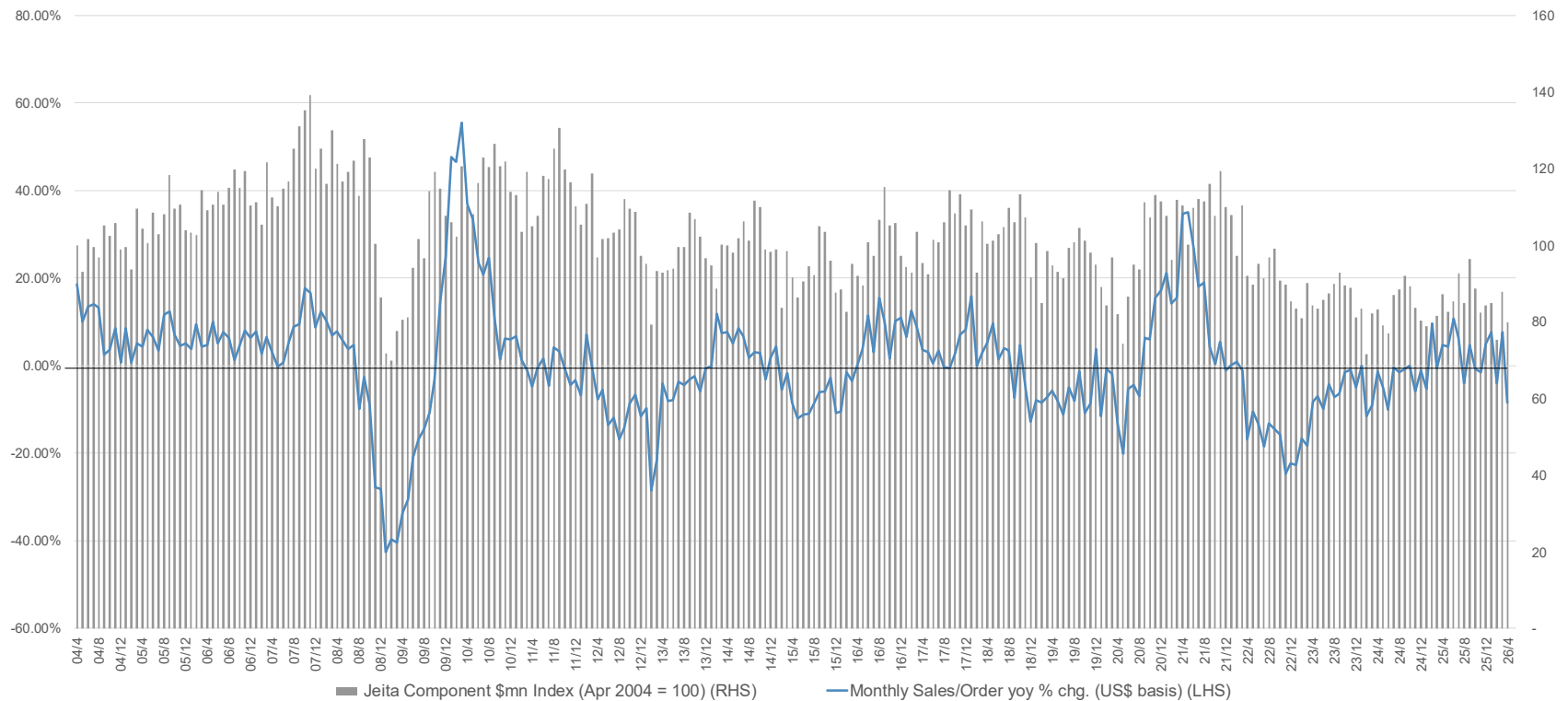
Source: JEITA, J.P. Morgan.

Electronic component market cap vs component monthly sales/orders



Note: Share prices as of 7/10/2025. Electronic component coverage market capitalization includes Murata Mfg., Taiyo Yuden, TDK, Kyocera, Rohm, Nichicon, Nippon Chemi-Con, Nidec, MinebeaMitsumi, Iriden, Hirose Electric, JAE, Alps Alpine, Niterra, Wacom, NISSHA.
Source: Bloomberg Finance L.P., JEITA, J.P. Morgan.

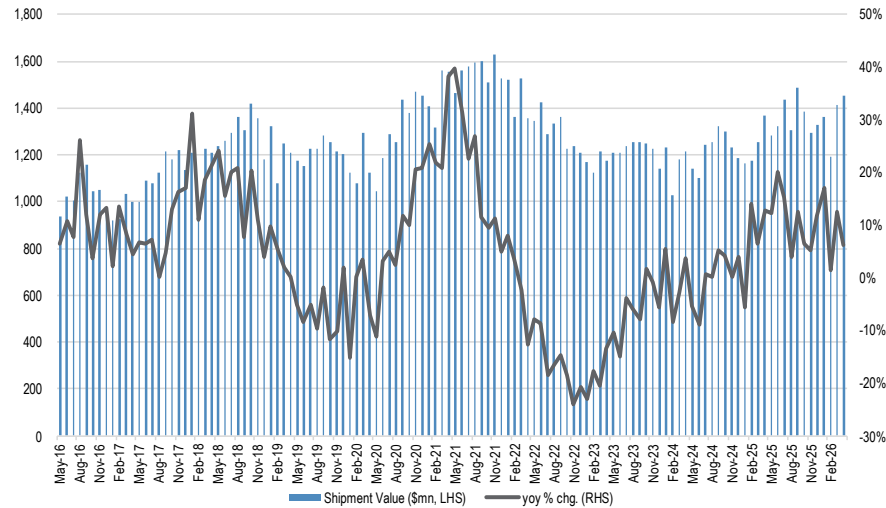
Electronic component sales index, yoy % chg.



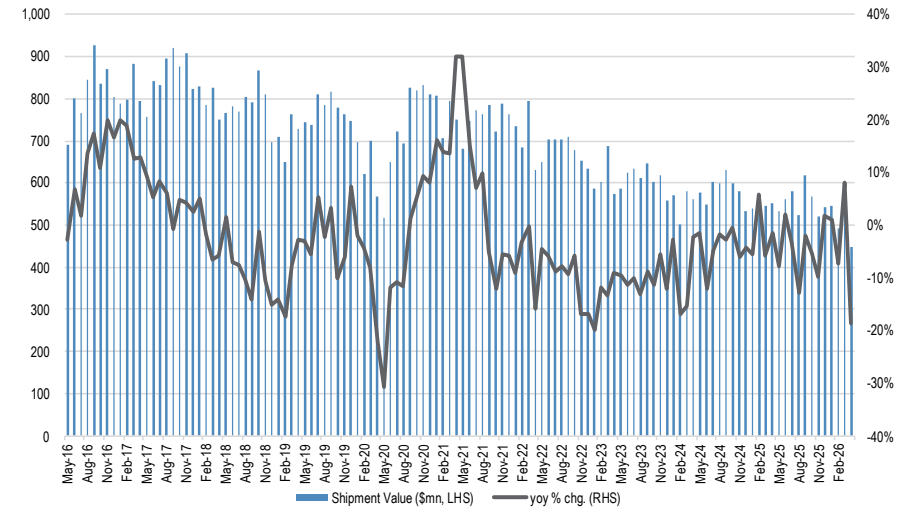
Source: Bloomberg Finance L.P., JEITA, J.P. Morgan.

JEITA global component shipments by category (\$mn)

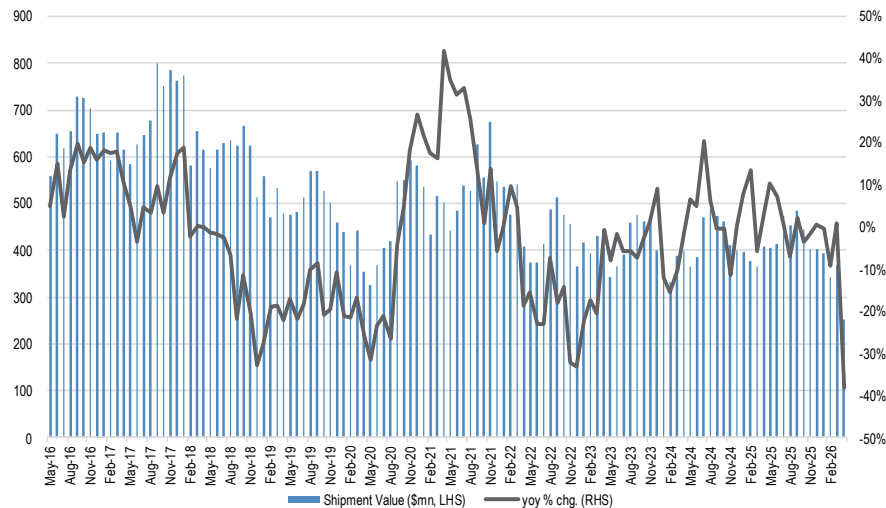
Passive components



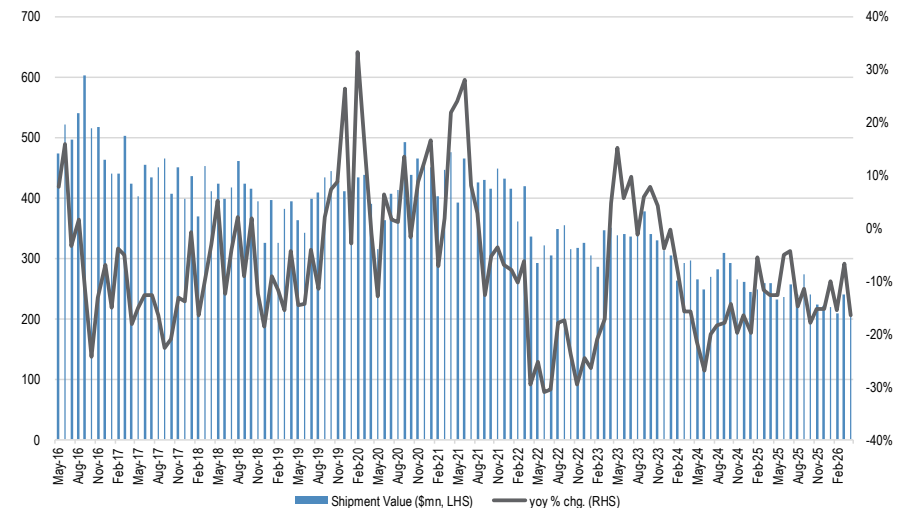
Connecting components



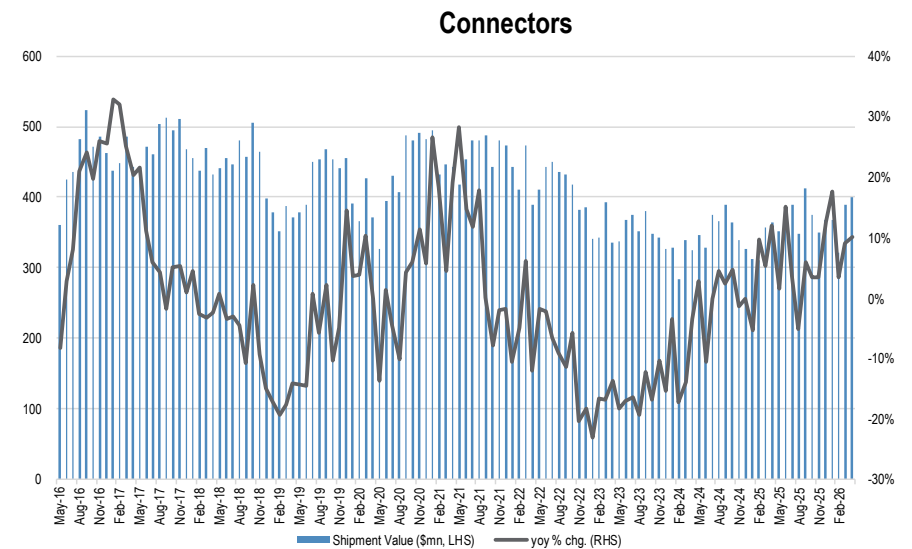
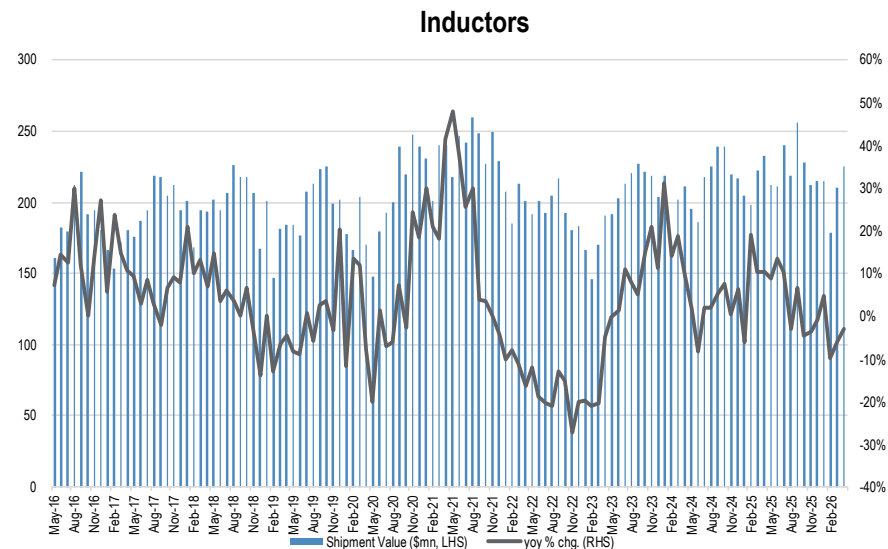
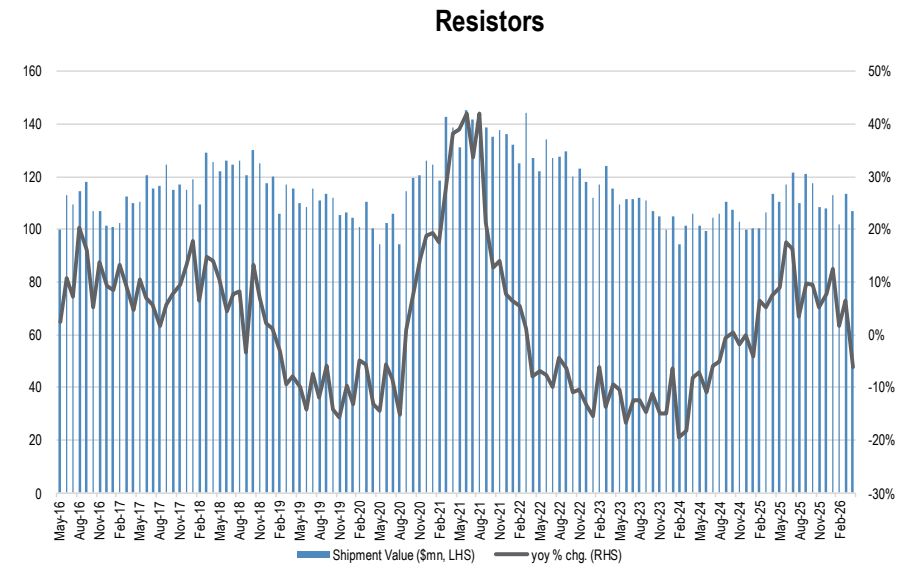
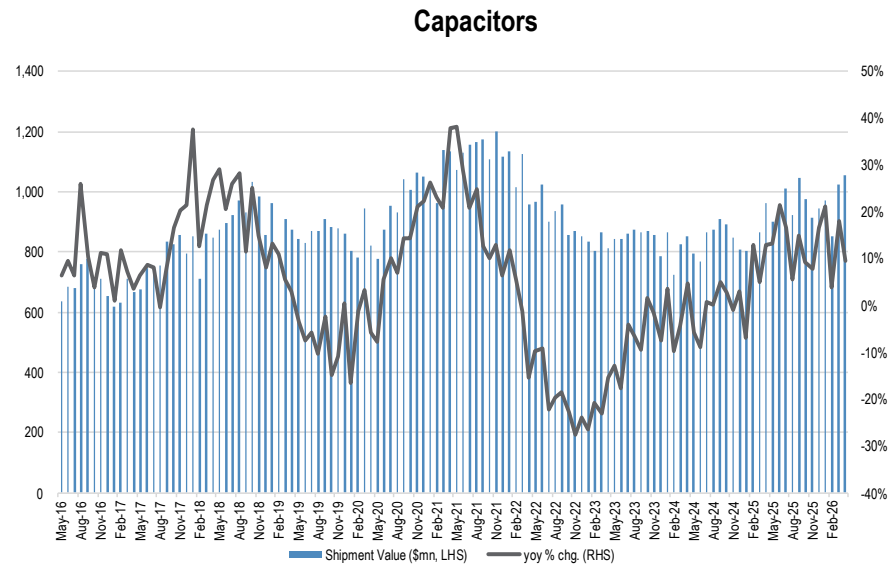
Transducers



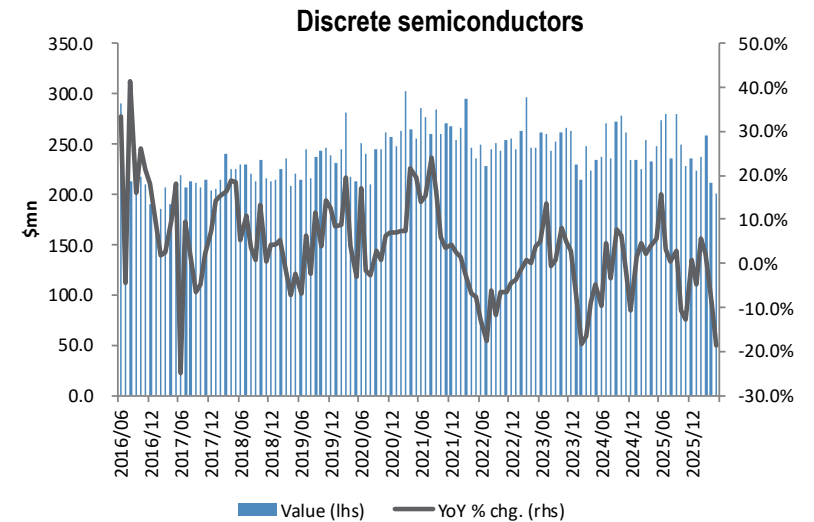
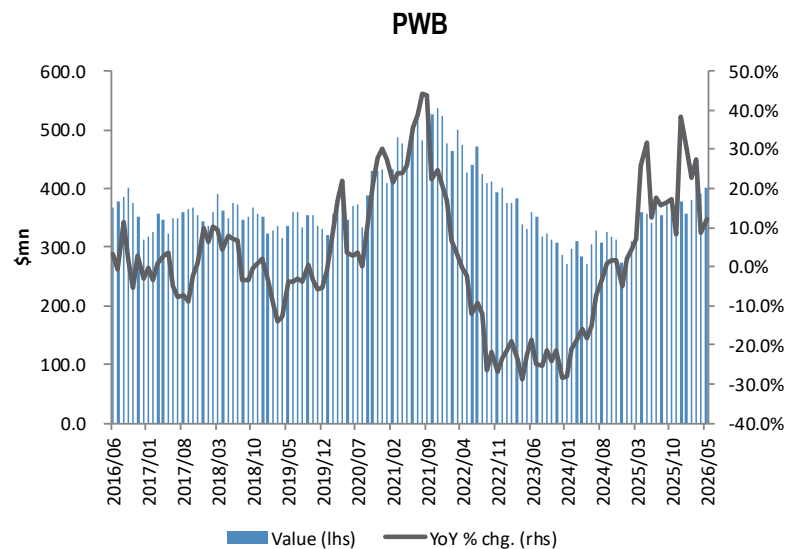
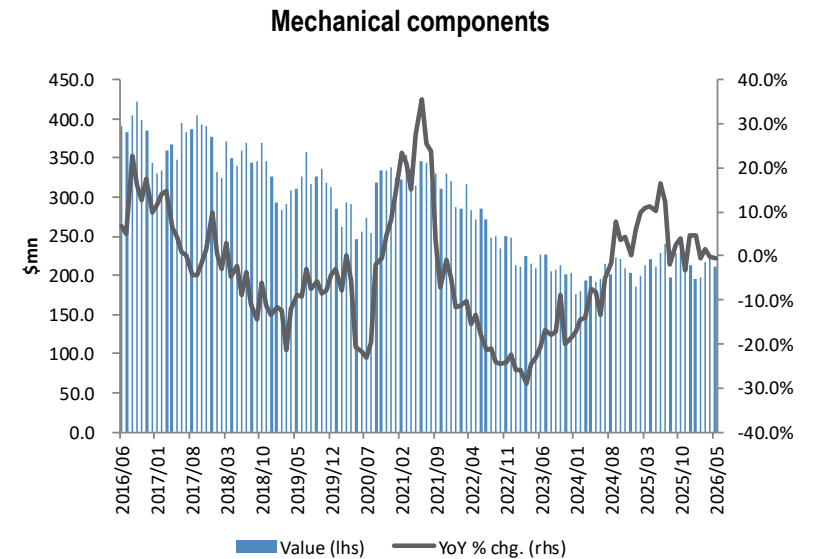
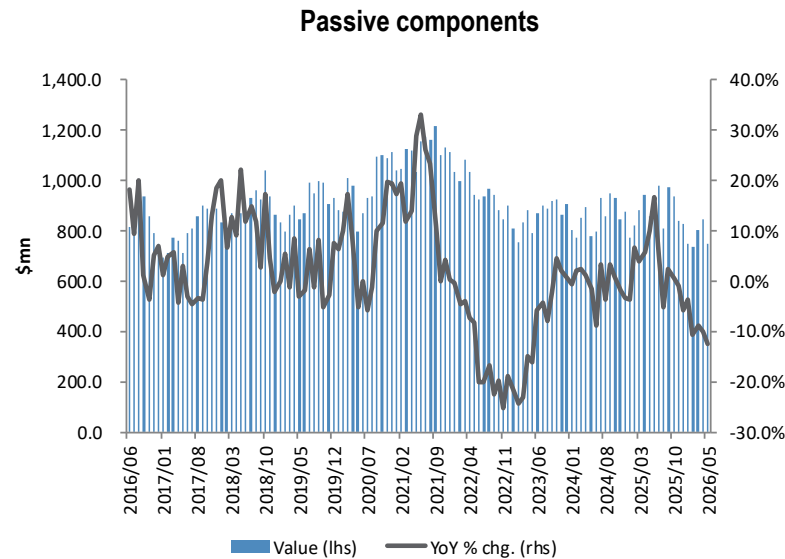
Other components



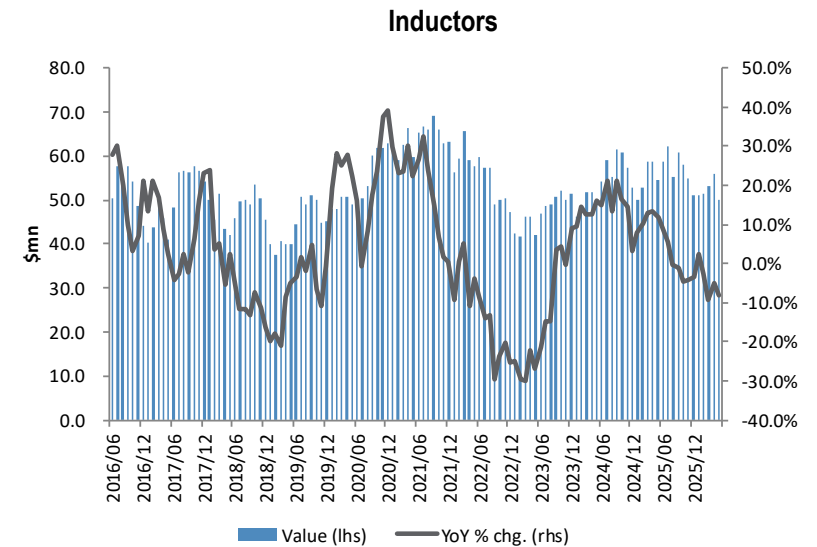
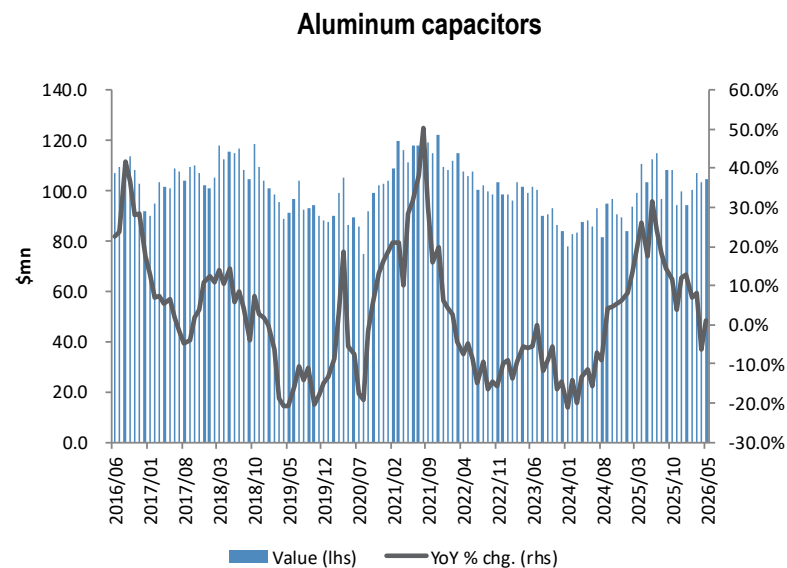
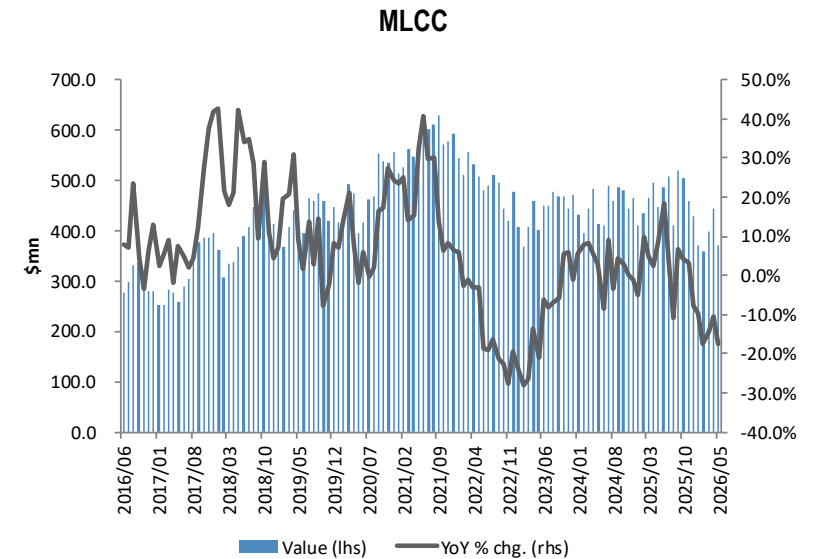
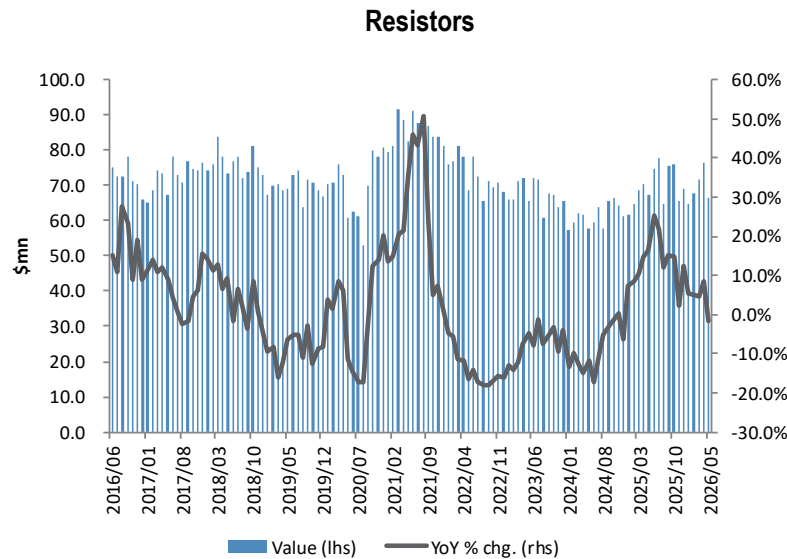
JEITA global component shipments by key component (\$mn)



Domestic component and semiconductor production value (\$mn)

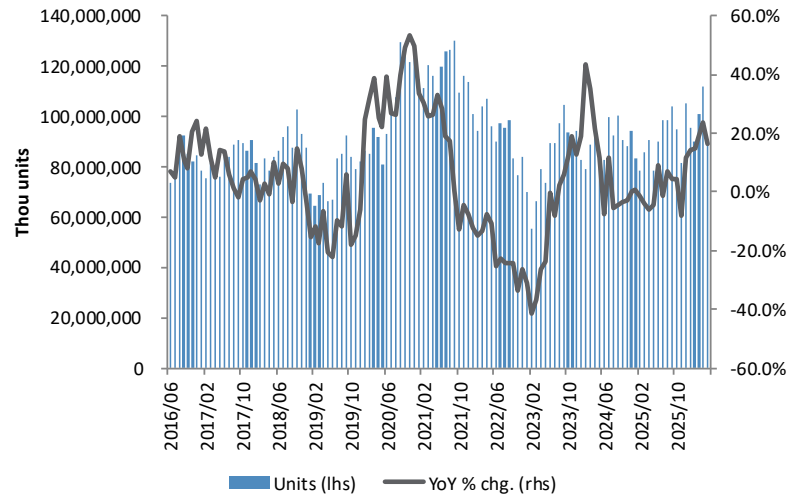


Domestic passive component production value (\$mn)

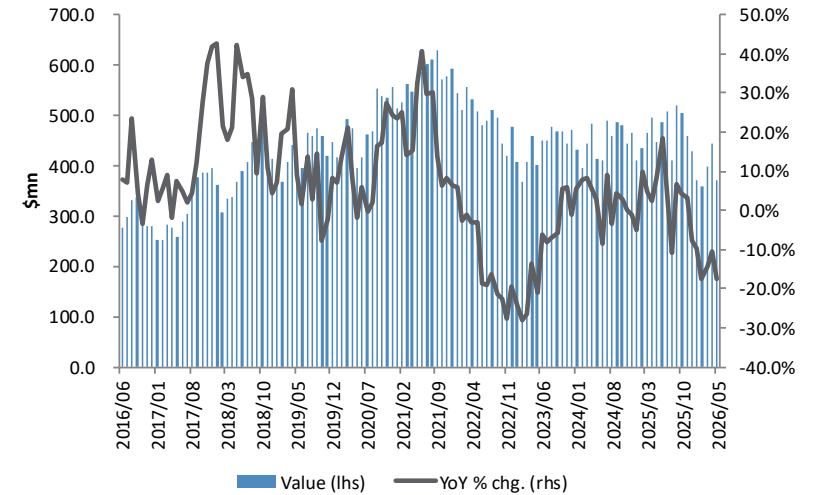


METI MLCC production volume, values and ASP in Japan

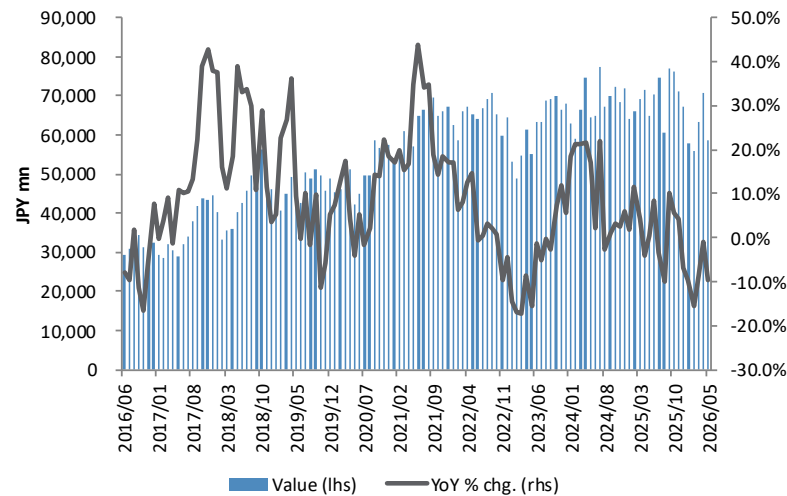
MLCC production volume in Japan



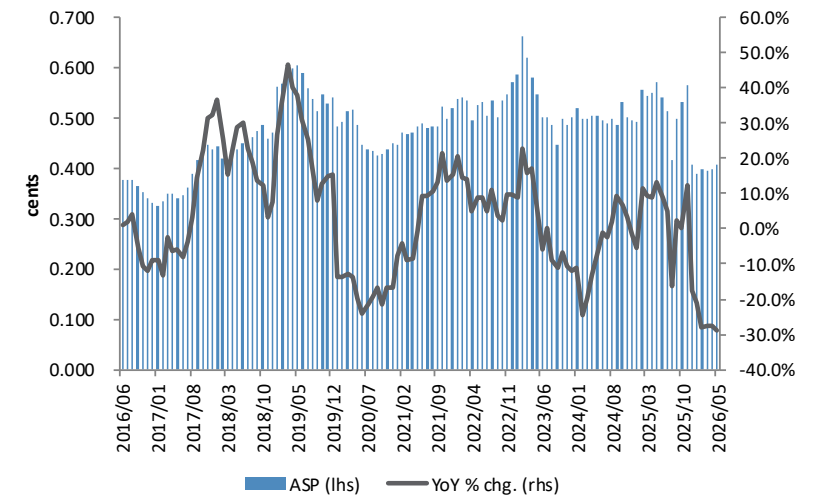
MLCC production value in Japan (\$ mn)



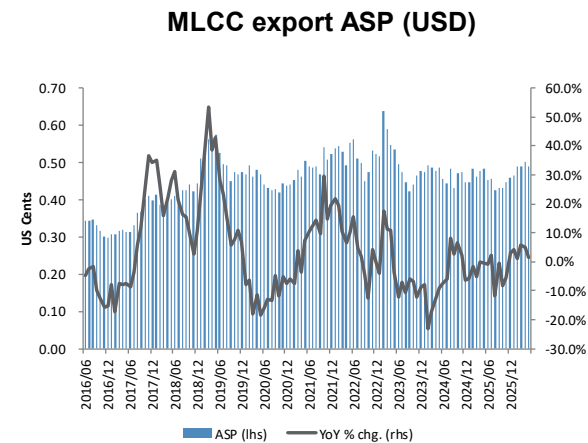
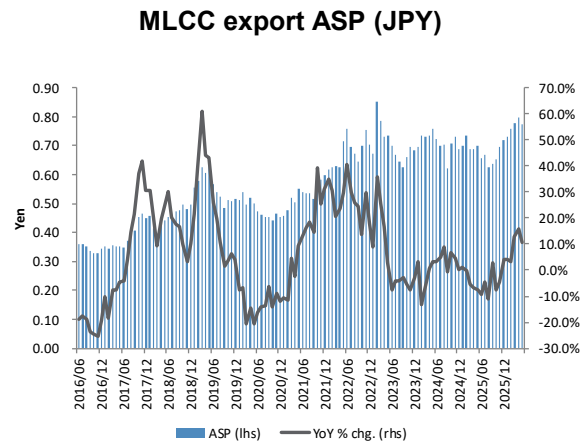
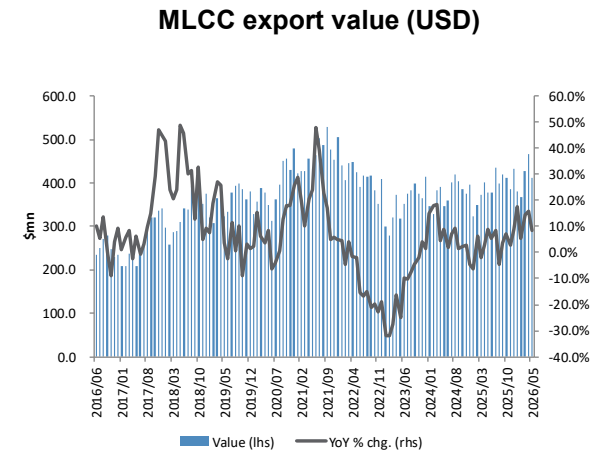
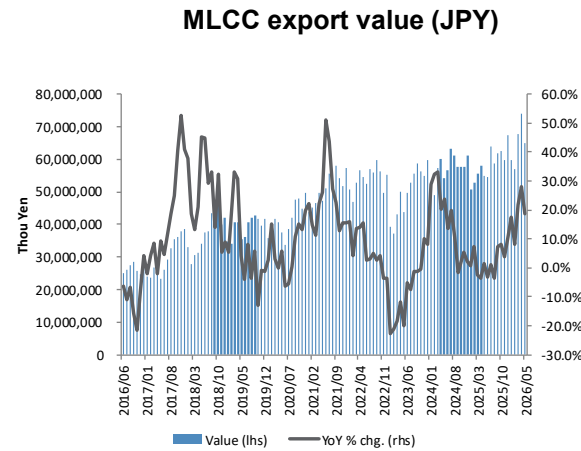
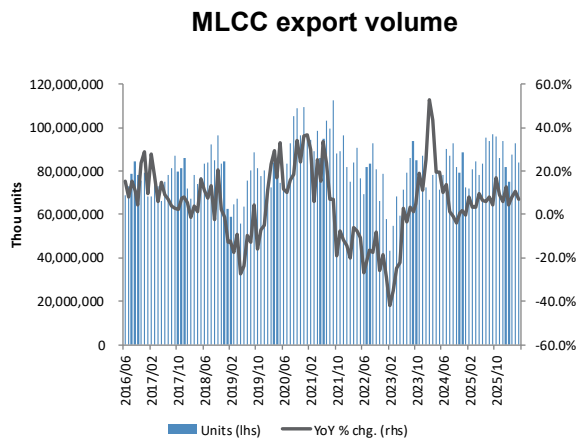
MLCC production value in Japan (JPY mn)



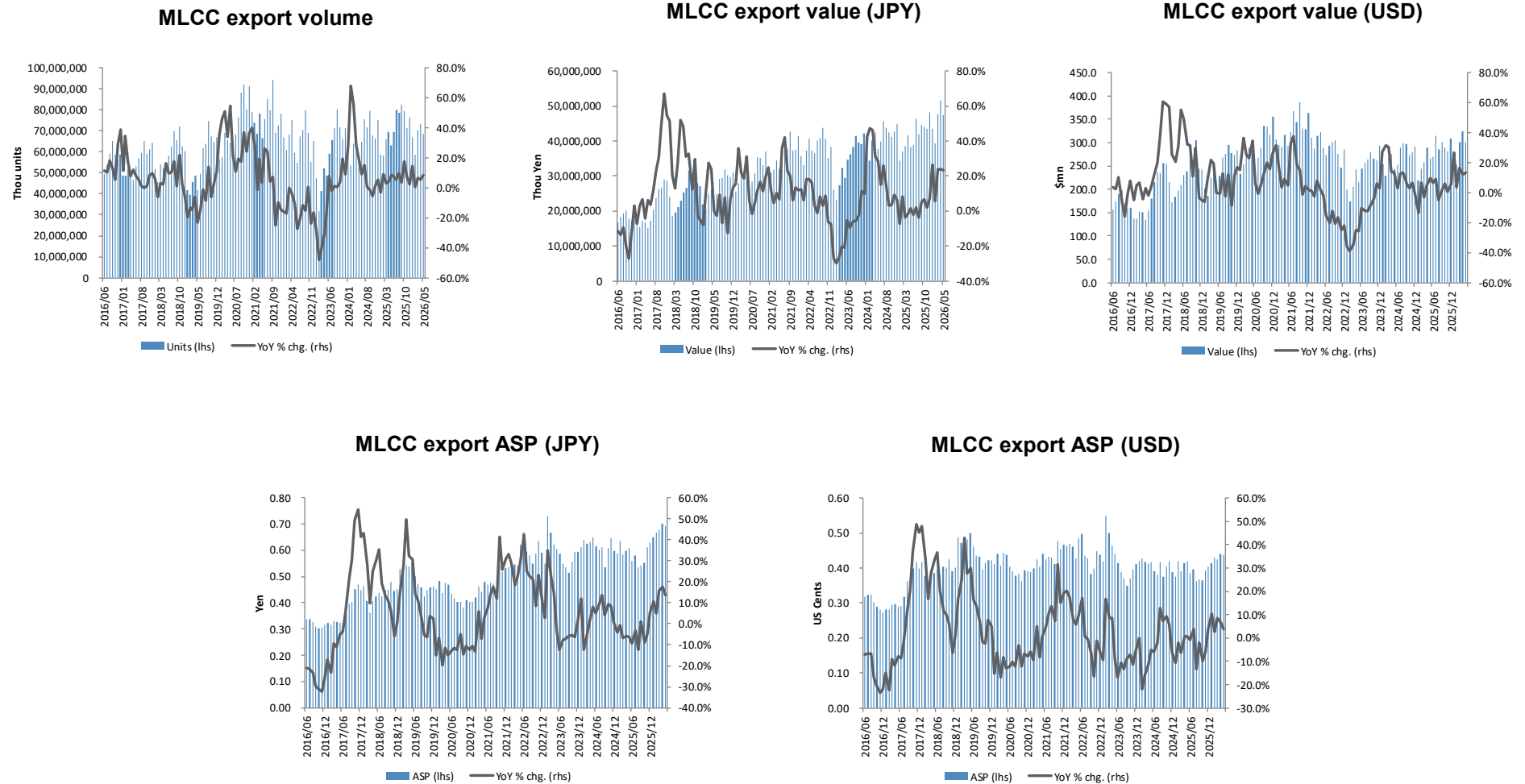
MLCC production ASP in Japan



MLCC export volume, value, and ASP from Japan



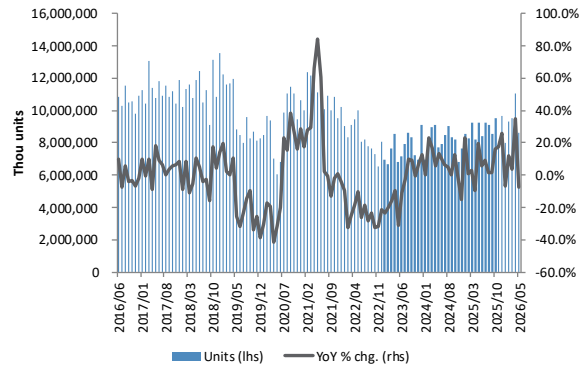
MLCC export volume, value and ASP from Japan to Korea, China, Taiwan, Hong Kong, Vietnam, and India markets



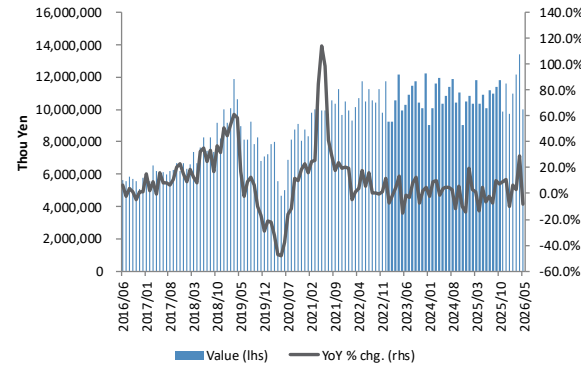
Source: MOF, J.P. Morgan.

MLCC export volume, value, and ASP from US and Europe

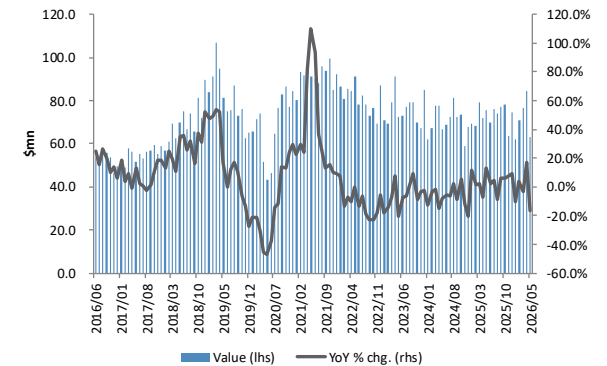
MLCC export volume



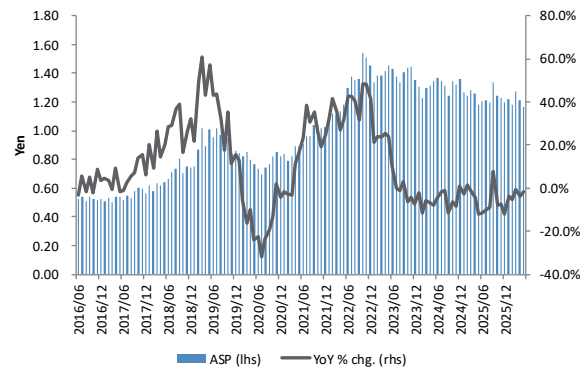
MLCC export value (JPY)



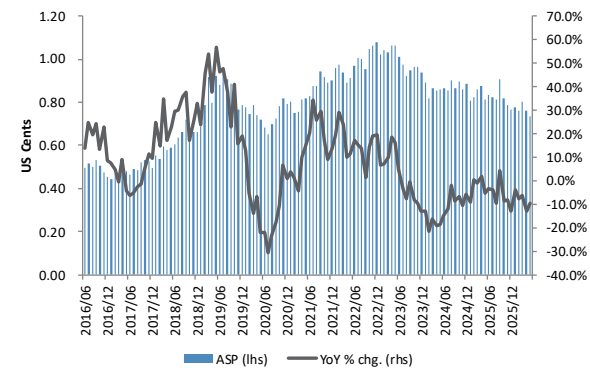
MLCC export value (USD)



MLCC export ASP (JPY)

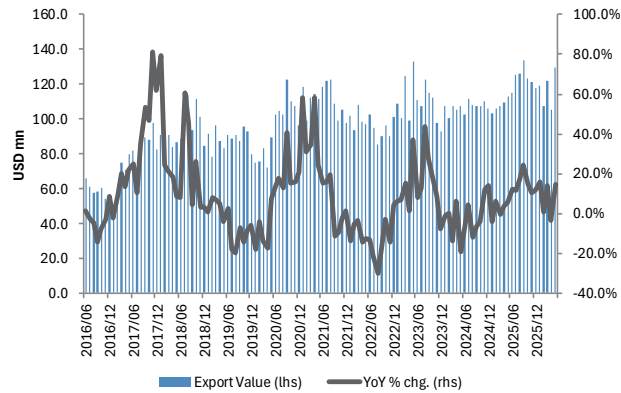


MLCC export ASP (USD)

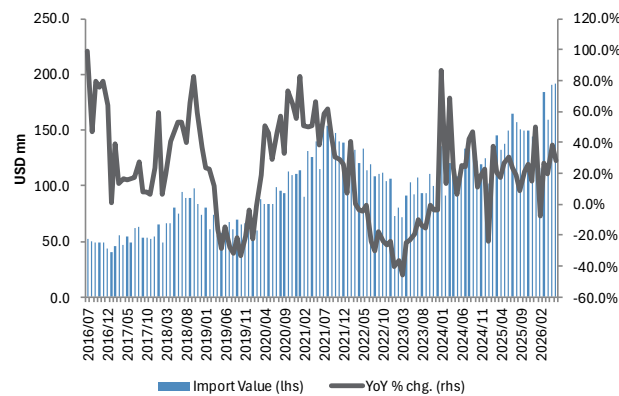


MLCC export / import value from Korea, Taiwan, China, Thailand and Malaysia to world

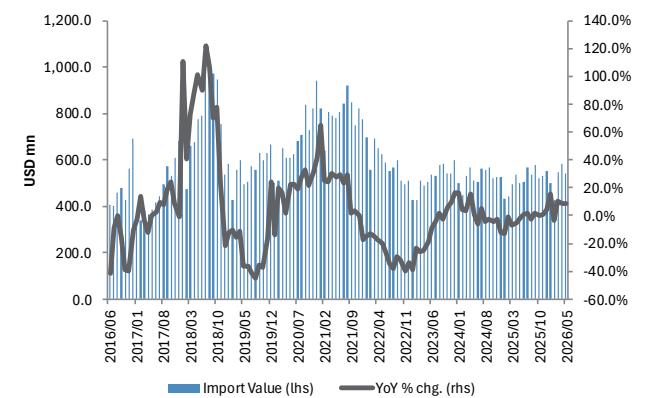
Korea : MLCC export value (USD)



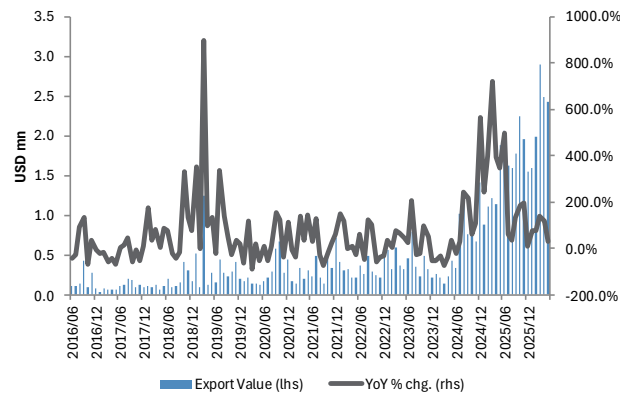
Taiwan : MLCC import value (USD)



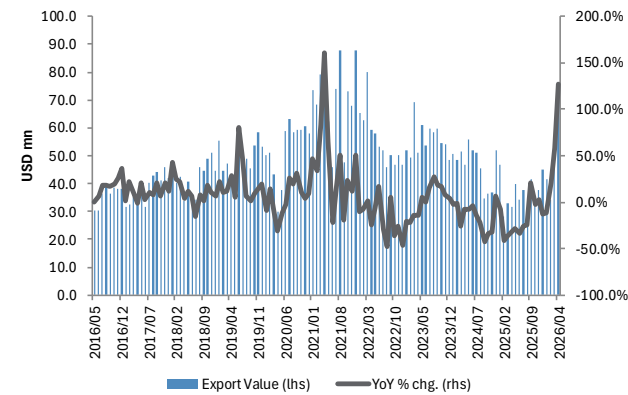
China : MLCC import value (USD)



Thailand : MLCC export value (USD)



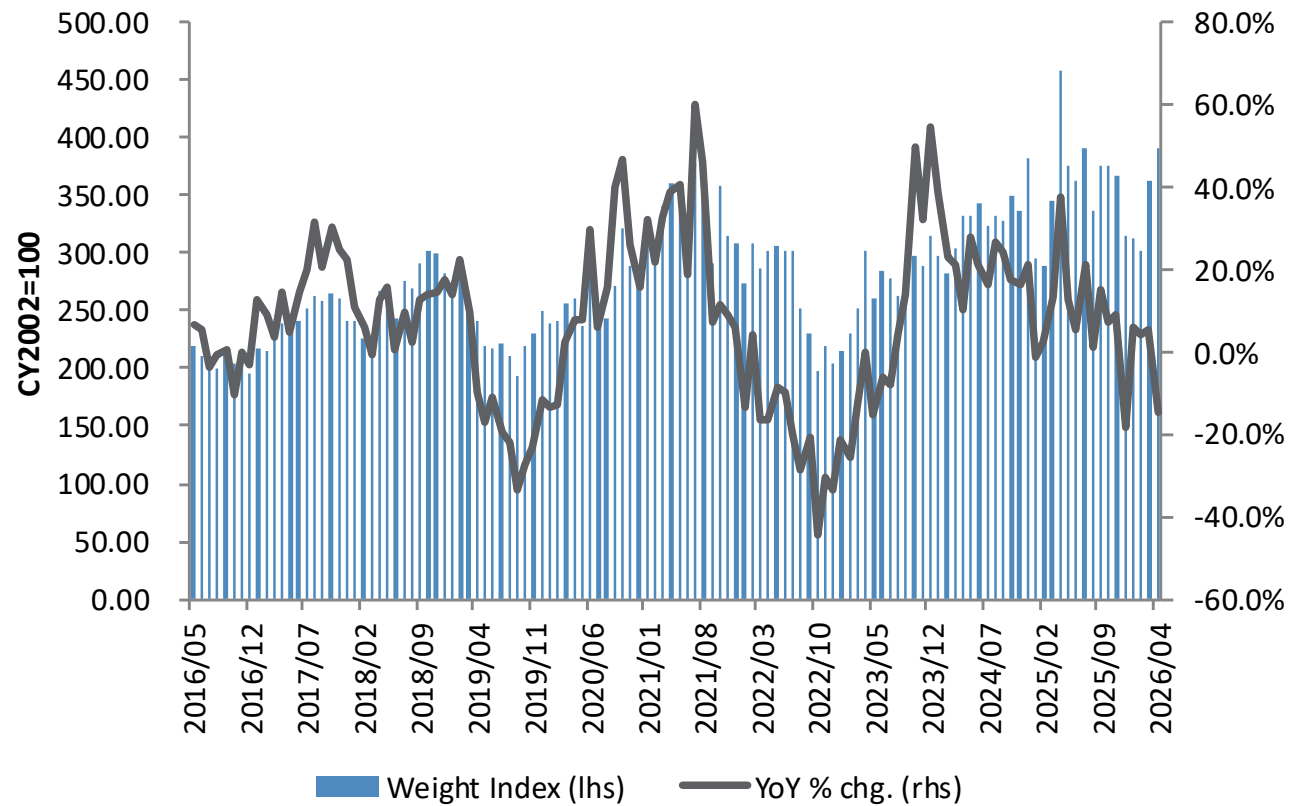
Malaysia : MLCC export value (USD)



Source: K-stat, CPT, Custom of China, Thailand's trade statistics, UN Comtrade database, J.P. Morgan.

Note: Korea: through Jun-26; Taiwan: through Jun-26 (May–Jun prelim.); China/Thailand: through May-26; Malaysia: through Apr-26.

Production volume of dielectric ceramic electrode materials



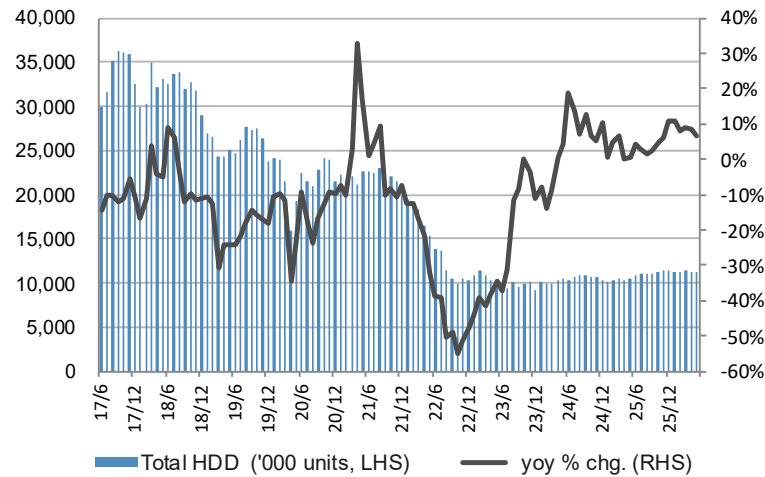
MLCC: Automotive MLCC contents per car

MLCC contents by powertrain

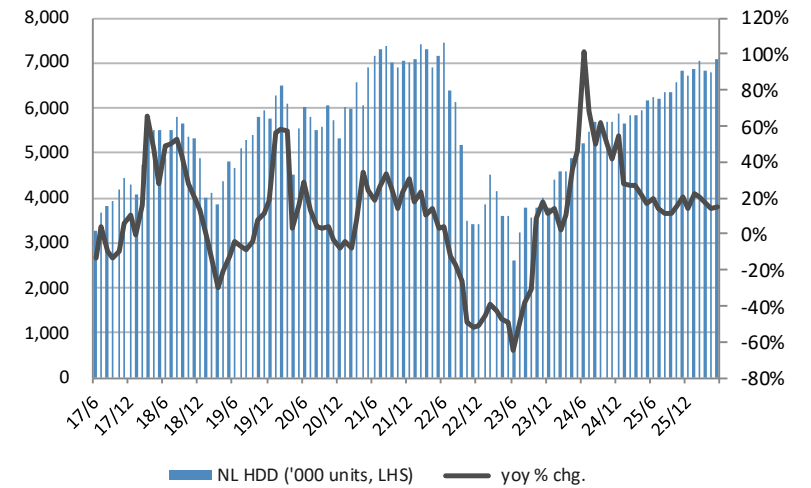
As of 2021					
	GAS	Mild HEV	Strong HV	PHEV	BEV
Power Train	300-500	-	1000-1600	1500-2000	2000-2500
ADAS	3000-5000				
Safety	300-1000				
Non Safety	500-2500				
Infortainment	500-2500				
Total	4600-11500	-	5300-12600	5800-13000	6300-13500
As of 2022					
	GAS	Mild HEV	Strong HV	PHEV	BEV
Power Train	300-500	-	1000-1200	1500-2000	2000-2500
ADAS	3000-5000				
Safety	300-1000				
Non Safety	500-2500				
Infortainment	500-2500				
Total	4600-11500	-	5300-12200	5800-13000	6300-13500
As of 2023					
	Pure ICE	Mild HV	Strong HV	PHEV	EV/FCV
Power Train	300-500	1000-1200	1200-1600	1500-2000	2000-2500
ADAS	3000-5000				
Safety	300-1000				
Non Safety	500-2500				
Infortainment	500-2500				
Total	4600-11500	5300-12200	5500-12600	5800-13000	6300-13500
As of 2025					
	Pure ICE	Mild HV	Strong HV	PHEV	EV/FCV
Power Train	300-500	1000-1200	1200-1600	2500-3000	2000-2500
ADAS	1500-5000 (L2/3 1500-3000, L4/5 3000-5000)				
Safety	300-1000				
Non Safety	500-2500				
Infortainment	500-2500				
Total	4600-11500	6500-16200	6700-16600	8000-18000	7500-17500

HDD monthly production volume and capacity

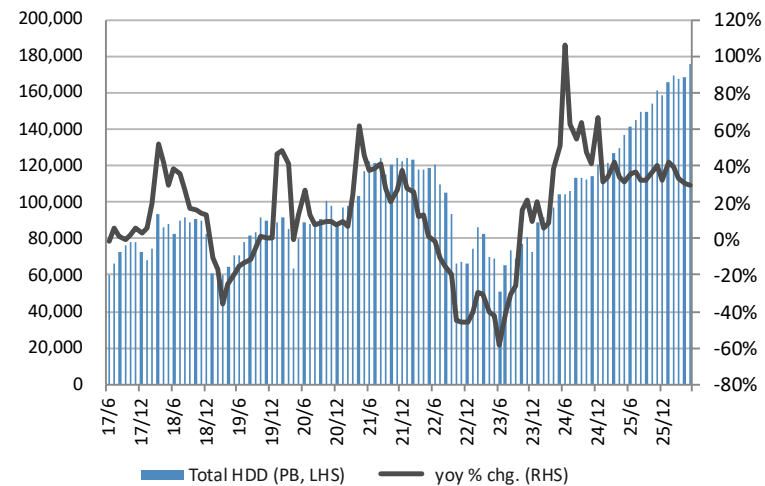
Total HDD production volume



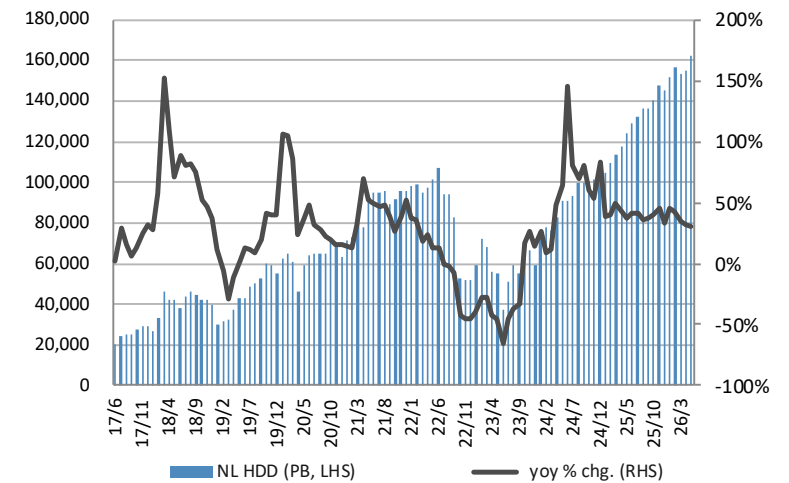
Nearline HDD production volume



Total HDD production capacity

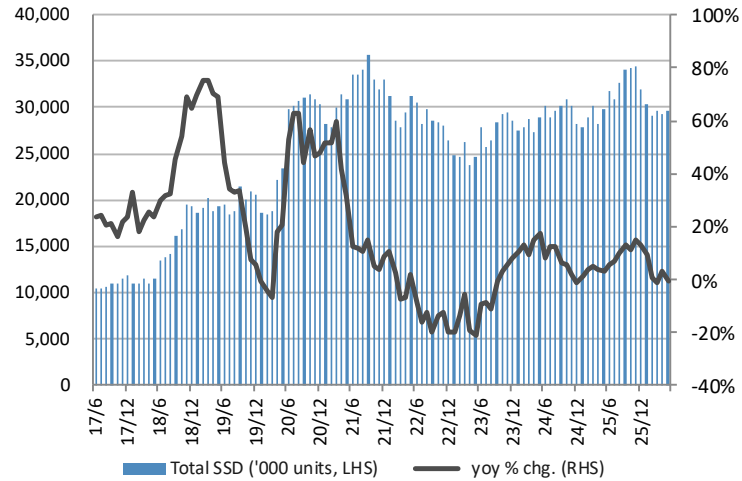


Nearline HDD production capacity

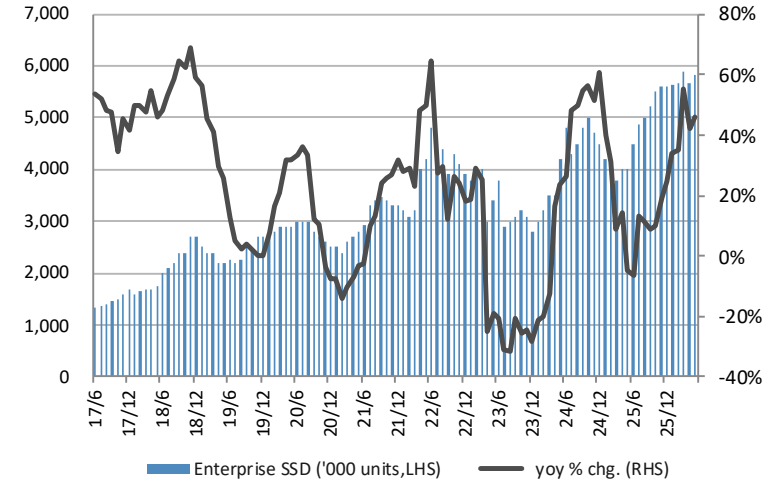


SSD monthly production volume and capacity

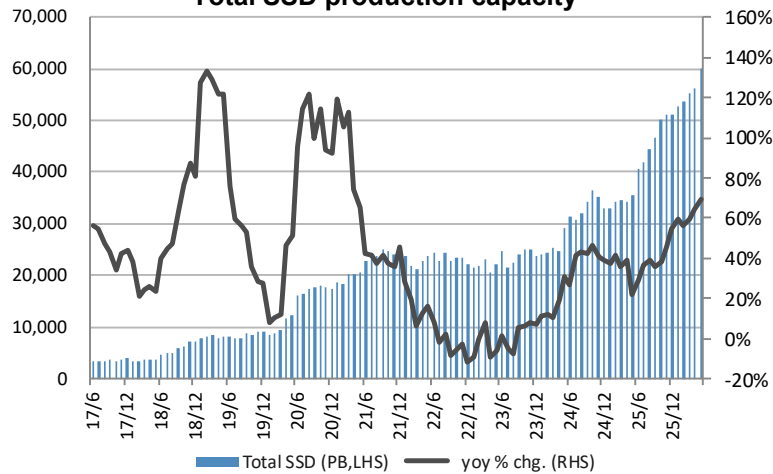
Total SSD production volume



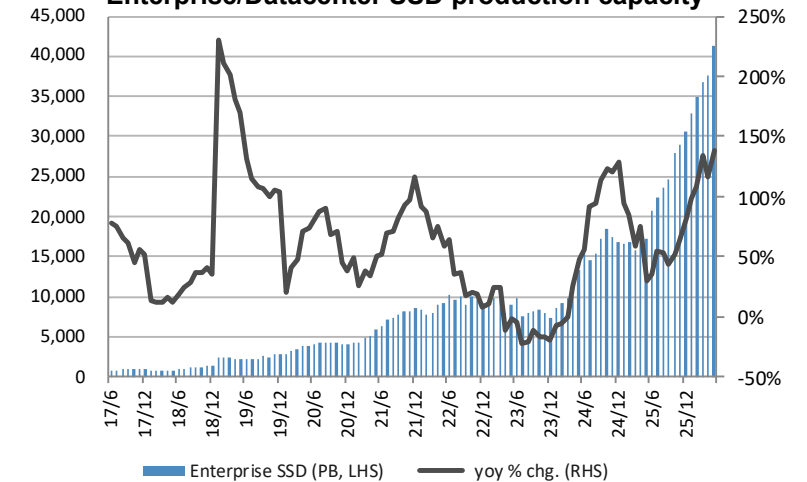
Enterprise/Datacenter SSD production volume



Total SSD production capacity

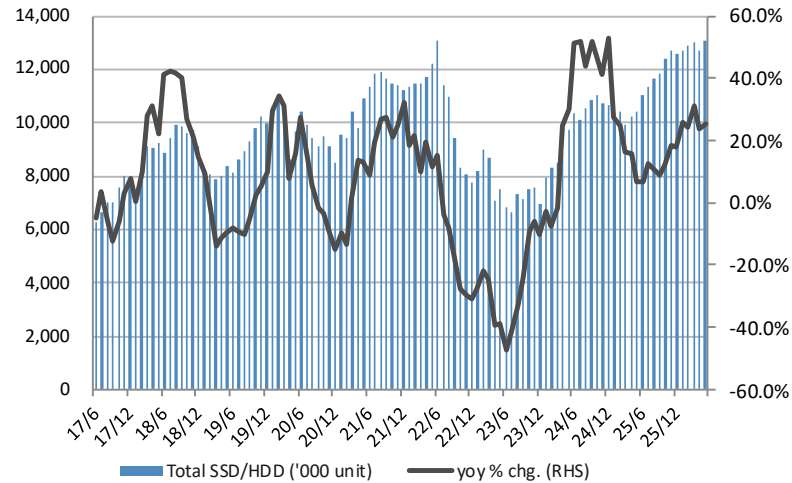


Enterprise/Datacenter SSD production capacity

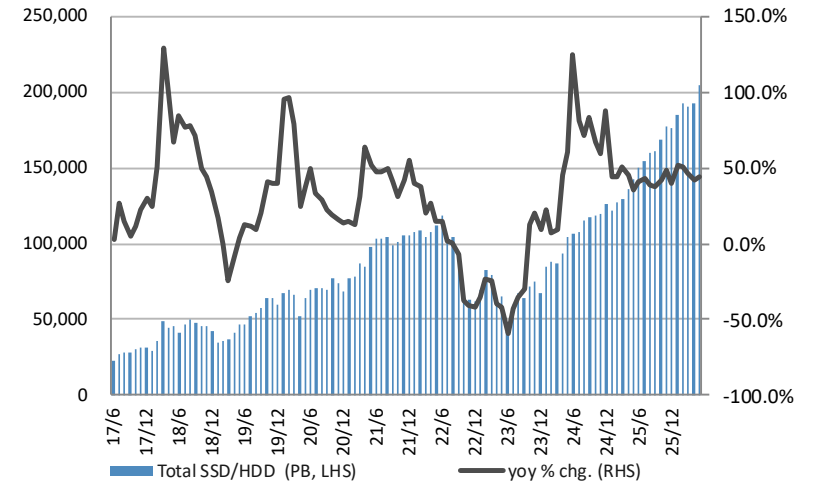


Enterprise/Datacenter storage production volume and capacity

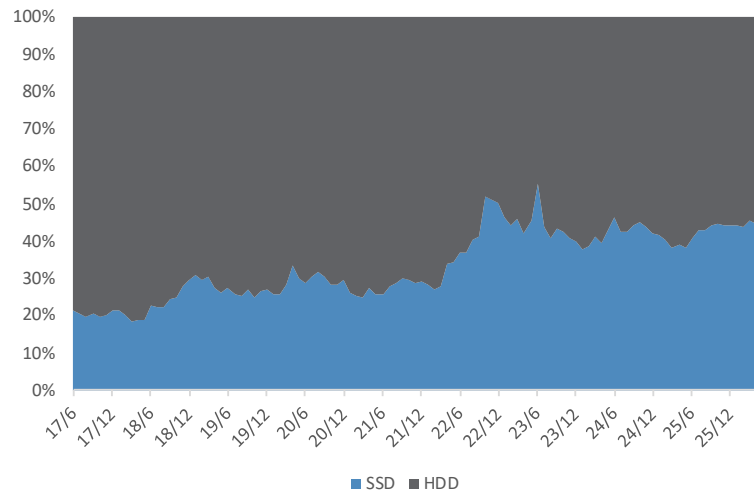
Enterprise/Datacenter HDD/SSD production volume



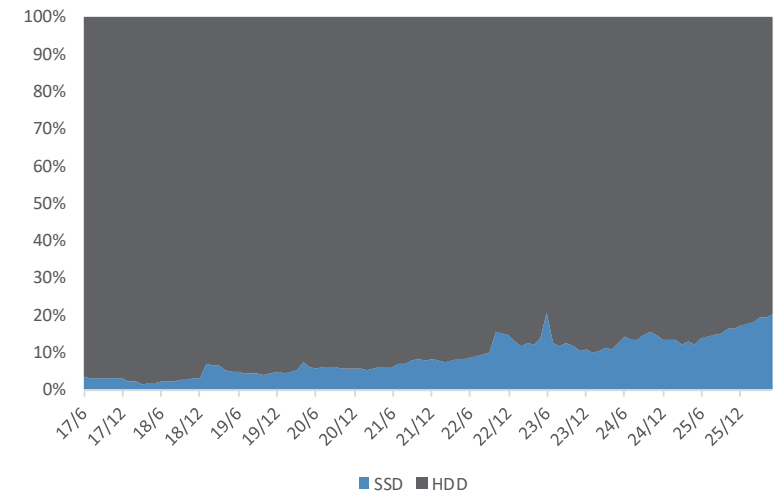
Enterprise/Datacenter HDD/SSD production capacity



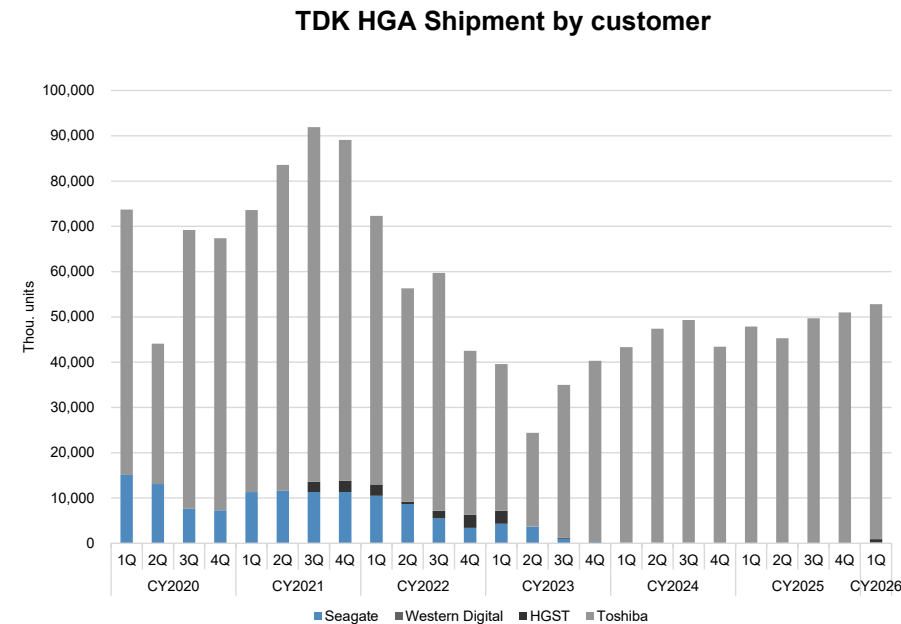
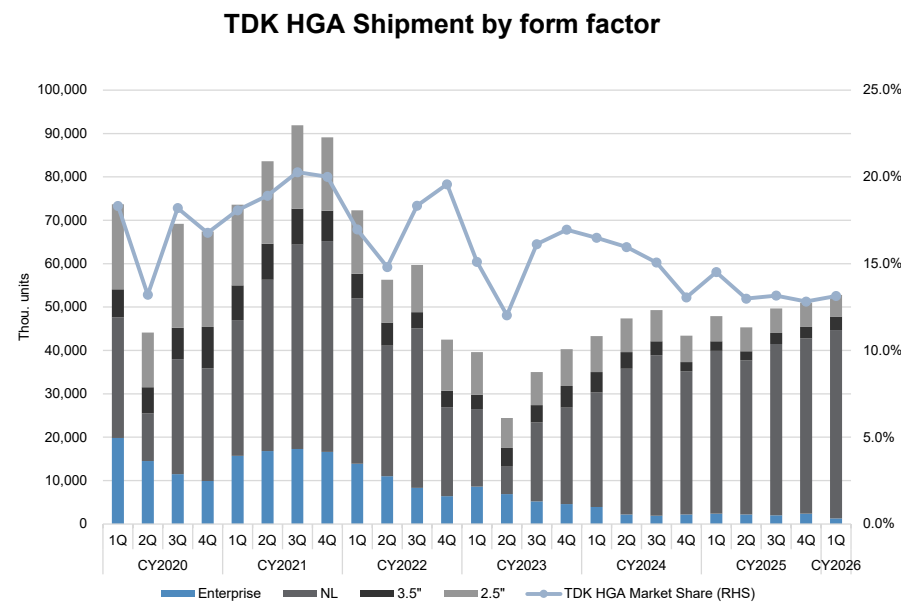
Enterprise/Datacenter HDD/SSD market share in volume



Enterprise/Datacenter HDD/SSD market share in capacity



TDK's HDD head market shipment and market share



Source: TSR, J.P. Morgan.

Discrete semiconductor market trend

Discrete Semiconductor demand by application

(\$mn)	2020	2021	2022	2023	2024	2025
Transistors						
Data Processing - Compute Electronics	1,285	1,555	1,620	1,344	1,483	1,776
Data Processing - Storage Electronics	115	131	167	133	118	121
Automotive Electronics	5,105	7,151	0	0	0	0
Industrial and Military/Civil Aerospace Electronics	5,203	6,876	7,865	8,126	6,977	7,399
Wireless Communications Electronics	2,079	2,334	2,372	1,749	1,540	1,519
Wired Communications Electronics	416	461	434	369	335	361
Consumer Electronics	2,228	2,951	3,043	2,749	2,526	2,604
Total	16,431	21,459	24,683	25,660	24,663	24,879
Diodes						
Data Processing - Compute Electronics	582	785	752	618	544	636
Data Processing - Storage Electronics	44	52	52	43	32	33
Automotive Electronics	1,268	1,666	1,412	1,192	1,157	1,200
Industrial and Military/Civil Aerospace Electronics	1,897	2,599	2,794	2,659	2,232	2,352
Wireless Communications Electronics	639	739	623	424	357	359
Wired Communications Electronics	205	233	227	199	165	173
Consumer Electronics	1,153	1,540	1,412	1,192	1,157	1,200
Total	5,788	7,614	7,724	7,148	6,639	6,908

(\$mn)	2020	2021	2022	2023	2024	2025
Total Discrete						
Data Processing - Compute Electronics	1,867	2,340	2,372	1,962	2,027	2,412
Data Processing - Storage Electronics	159	183	219	176	150	154
Automotive Electronics	6,373	8,817	1,412	1,192	1,157	1,200
Industrial and Military/Civil Aerospace Electronics	7,100	9,475	10,659	10,785	9,209	9,751
Wireless Communications Electronics	2,718	3,073	2,995	2,173	1,897	1,878
Wired Communications Electronics	621	694	661	568	500	534
Consumer Electronics	3,381	4,491	4,455	3,941	3,683	3,804
Total	22,219	29,073	32,407	32,808	31,302	31,787

Discrete Semiconductor market share by application

Transistors		All Market					
2024	2025		2024	2025	2024	2025	2025
#	#		Sales (mn\$)	Sales (mn\$)	Share	Share	YoY Change
1	1	Infineon Technologies	4,581	4,487	18.6%	18.0%	-2.1%
2	2	Onsemi	2,727	2,351	11.1%	9.4%	-13.8%
4	3	DENSO	1,262	1,384	5.1%	5.6%	9.7%
5	4	Fuji Electric	1,107	1,160	4.5%	4.7%	4.8%
3	5	STMicroelectronics	1,783	1,138	7.2%	4.6%	-36.2%
6	6	Nexperia	1,087	1,062	4.4%	4.3%	-2.3%
8	7	Toshiba	801	882	3.2%	3.5%	10.1%
7	8	Mitsubishi Electric	861	809	3.5%	3.3%	-6.0%
9	9	Rohm	724	785	2.9%	3.2%	8.4%
-	10	Hangzhou Silan Microelectronics	553	667	2.2%	2.7%	20.6%
10	-	Ampleon	644	662	2.6%	2.7%	2.8%
Top 10			15,577	14,725	63.2%	59.2%	-5.5%
All Others			9,086	10,154	36.8%	40.8%	11.8%
Total Market			24,663	24,879	100.0%	100.0%	0.9%

Transistors Data Processing - Compute Electronics

2024 #	2025 #		2024 Sales (mn\$)	2025 Sales (mn\$)	2024 Share	2025 Share	2025 YoY Change
1	1	Infineon Technologies	403	469	27.2%	26.4%	16.4%
2	2	Alpha and Omega Semiconductor	169	176	11.4%	9.9%	4.1%
3	3	Onsemi	136	158	9.2%	8.9%	16.2%
4	4	Toshiba	81	121	5.5%	6.8%	49.4%
5	5	Nexperia	76	85	5.1%	4.8%	11.8%
6	6	Diodes	68	80	4.6%	4.5%	17.6%
7	7	Rohm	67	71	4.5%	4.0%	6.0%
8	8	Texas Instruments	37	48	2.5%	2.7%	29.7%
-	9	Yangjie Electronic	29	42	2.0%	2.4%	44.8%
10	10	Littelfuse	32	40	2.2%	2.3%	25.0%
9	-	Vishay	34	37	2.3%	2.1%	8.8%
-	-	uPI Semi	21	36	1.4%	2.0%	71.4%
Top 10			1,103	1,290	74.4%	72.6%	17.0%
All Others			380	486	25.6%	27.4%	27.9%
Total Market			1,483	1,776	100.0%	100.0%	19.8%

Rohm, Toshiba Semiconductor, MELCO semiconductor fab locations

ROHM Production Sites	Discrete	Front-end Process		Back-end Process
		Analog, Logic	Others	
Head Office Plant	X	X	X	
Shiga Plant	X			
Hamamatsu Plant		X	X	
Okayama Plant (ROHM Wako Co., Ltd.)		X	X	X
Hirokawa Head Office Plant (ROHM Apollo Co., Ltd.)	X		X	X
Chikugo Plant (ROHM Apollo Co., Ltd.)	X			
Yukuhashi Plant (ROHM Apollo Co., Ltd.)		X		
Miyazaki Plant (LAPIS Semiconductor Co., Ltd.)	X	X		
Miyazaki No. 2 Plant (LAPIS Semiconductor Co., Ltd.)	X			
Miyagi Plant (LAPIS Semiconductor Co., Ltd.)		X		
ROHM Mechatech Co., Ltd.				X
Philippines Plant			X	X
Thailand Plant				X
Malaysia Plant				X
Korea Plant				X
Tianjin Plant				X
Dalian Plant				X
Toshiba Semiconductor Business	Discrete	Front-end Process		Back-end Process
		Analog, Logic	Others	
Kaga Plant	X			
Himeji Plant	X			X
Kitakami Plant (Japan Semiconductor Corporation)		X		
Oita Plant (Japan Semiconductor Corporation)		X		
Buzen Plant				X
Thailand Plant				X
Mitsubishi Electric (Discrete Semiconductors Only)	Discrete	Front-end Process		Back-end Process
		Analog, Logic	Others	
Fukuyama Plant	X			X
Kumamoto Plant (Koshi City)	X			
Shisui Plant	X			
Itami Plant	X			
Fukuoka Plant				X
Fukuoka Plant (MELCO Power Devices)				X
Nijo Plant (MELCO Power Devices)				X
Hikami Plant (MELCO Power Devices)				X
Toyooka Plant (MELCO Power Devices)				X
Shimane Plant (MELCO Power Devices)				X
Hefei Plant (China, MELCO Power Devices)				X
Hungary Plant (Vincotech Hungary)				X

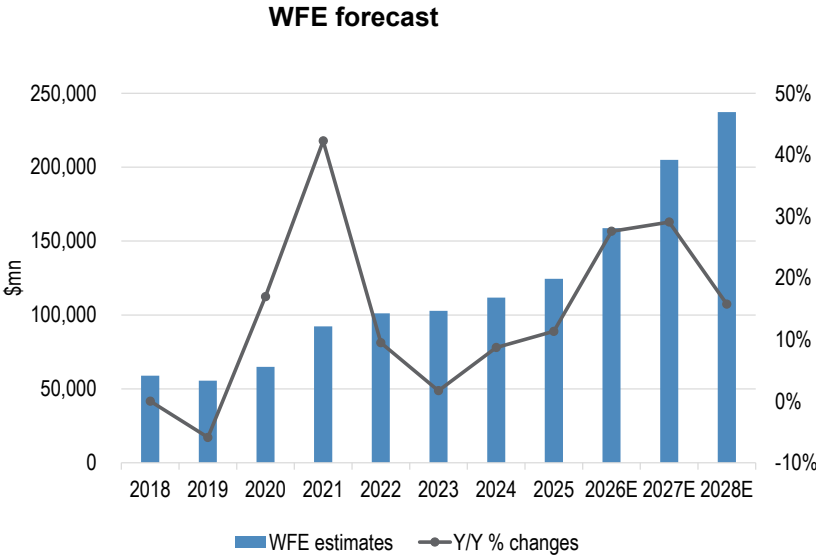
Source: Company data.

AI server shipment forecasts

k units	2022	2023	2024	2025	2026E	2027E	2028E	25-28 CAGR
Total datacenter AI chip volumes	2,153	3,632	7,815	10,320	16,045	24,093	28,234	40%
YoY growth		69%	115%	32%	55%	50%	17%	
High end datacenter GPU volumes	800	1,931	4,950	6,150	8,125	9,710	10,701	20%
YoY growth		141%	156%	24%	32%	20%	10%	
Nvidia	775	1,911	4,500	5,500	7,425	8,910	9,801	21%
YoY growth		147%	135%	22%	35%	20%	10%	
Arm-based			126	3,600	5,940	7,128	8,331	
Arm-based %			3%	65%	80%	80%	85%	
x86-based	775	1,911	4,374	1,900	1,485	1,782	1,470	
AMD	25	20	450	650	700	800	900	11%
YoY growth		-20%	2150%	44%	8%	14%	13%	
AI GPU peripheral cards					300	2,133	2,833	
Nvidia CPX						633	833	
Nvidia LPU					300	1,500	2,000	
ASIC chip volume	1,353	1,701	2,865	4,170	7,620	12,250	14,700	52%
YoY growth		26%	68%	46%	83%	61%	20%	
Chip per server								
Chip per ASIC server	5	5	7	7	8	8	8	
GPU per server								
GB structure			4	4	4	4	4	
HGX structure	8	8	8	8	8	8	8	
LPU structure	16	16	16	16	16	16	16	
Total AI server volumes	401	577	1,066	1,823	2,689	3,589	4,237	32%
YoY growth		44%	85%	71%	47%	33%	18%	
High end GPU AI server shipment	100	241	635	1,219	1,777	2,119	2,400	25%
YoY growth		141%	163%	92%	46%	19%	13%	
AI ASIC server shipment	301	336	432	605	912	1,470	1,838	45%
YoY growth		12%	28%	40%	51%	61%	25%	
Global server shipment (mn units)	13.8	11.4	12.1	12.1	15.0	17.8	19.9	18%
Total AI server % mix	2.9%	5.1%	8.8%	15.0%	17.9%	20.2%	21.3%	
High end GPU AI server % mix	0.7%	2.1%	5.3%	10.1%	11.8%	11.9%	12.1%	
ASIC chip per servers	2.2%	3.0%	3.6%	5.0%	6.1%	8.3%	9.3%	

Note: Our Taiwan technology team's forecasts as of June 2026
Source: Company reports, Gartner®, J.P. Morgan estimates

WFE forecast



Semi capex forecast

	2017	2018	2019	2020	2021	2022	2023	2024E	2025E	2026E	2027E
Memory	42,500	54,727	45,584	44,596	57,329	64,233	53,653	60,751	77,307	114,866	185,789
NAND	28,570	29,891	25,849	25,197	28,849	30,910	22,804	15,634	18,081	23,434	34,232
DRAM	13,930	24,836	19,735	19,399	28,479	33,323	30,849	45,117	59,226	91,432	151,557
Foundry	21,896	17,211	24,114	33,027	49,820	59,502	56,786	50,751	59,109	81,771	94,009
Logic	14,950	17,880	20,150	18,589	27,000	38,100	38,750	37,020	30,000	30,250	36,600
Total	79,346	89,818	89,849	96,212	134,149	161,835	149,189	148,522	166,416	226,887	316,398

	2018	2019	2020	2021	2022	2023	2024E	2025E	2026E	2027E
Memory	29%	-17%	-2%	29%	12%	-16%	13%	27%	49%	62%
NAND	5%	-14%	-3%	14%	7%	-26%	-31%	16%	30%	46%
DRAM	78%	-21%	-2%	47%	17%	-7%	46%	31%	54%	66%
Foundry	-21%	40%	37%	51%	19%	-5%	-11%	16%	38%	15%
Logic	20%	13%	-8%	45%	41%	2%	-4%	-19%	1%	21%
Total	13%	0%	7%	39%	21%	-8%	0%	12%	36%	39%

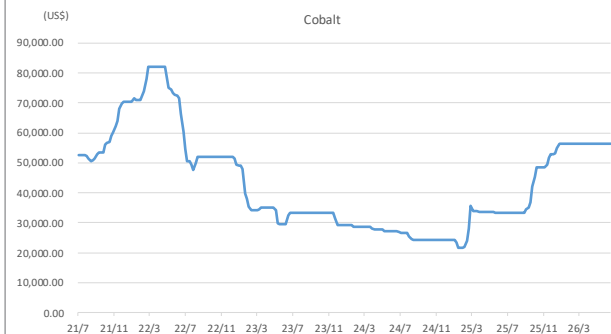
Apple RF FEM suppliers

	Product	Maker	iPhone15	iPhone16	iPhone16e	iPhone17	iPhone Air	iPhone 18 Pro/ Pro Max/Foldable	iPhone 18 /Air/ New SE	New iPad
FEM	Coupler Module	Skyworks	2	2	4	2	4			
	Low-band PAMiD	Skyworks	1	1	1	1	1			1
	Middle/High-band PAMiD	Broadcom	1	1	1	1	1			
	PAMiF	Broadcom	1	1	1	1	1			
Sub FEM	5G RX Module	Broadcom	1	1	x	1	x			
		Skyworks			1		1			
	A-Multiplexer	Broadcom	3	3	3	1	x			
		MURATA	1	1		x	1	1	1	
		Skyworks			1	1	2			
		Qualcomm				1				
		Qorvo				1	1			
	Diversity FEM MIMO	MURATA			1		1	0.5	0.5	
		Qualcomm		2		2				
		Skyworks	2	x	1	x	1			
	Diversity FEM Low-band	MURATA	1	1	1	x	x		1	
		Qualcomm				1				
		Skyworks					1			
	EN-DC FEM	Broadcom			1		1			
		Skyworks	1	1		1				
NCW	GPS FEM	Skyworks	1	1	1	1	1			
	WiFi 2.4GHz FEM	MURATA	2		2	2	2			
		Skyworks		2						
BAW Filter	WiFi 5GHz FEM	Skyworks	2	2	2	2	2			
		Broadcom	1	2	4	1	1			
	WiFi Filter (5GHz/5-7GHz)	Qorvo	2			1	1			
Murata's Dollar contents			3	2.5	2.5	1.5		1.5	3.5	2.5

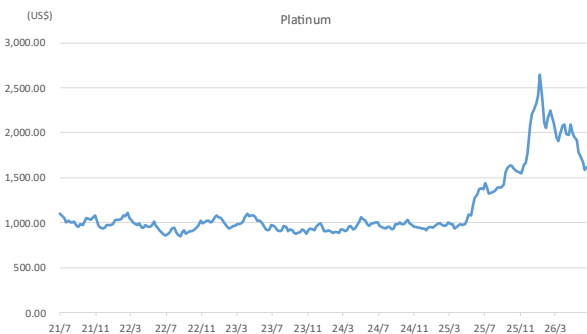
Note: All J.P. Morgan estimates are as of Dec 2025.
Source: Navian, J.P. Morgan estimates

Material price trend

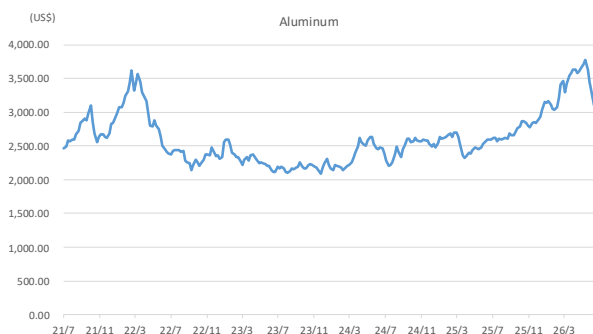
Cobalt



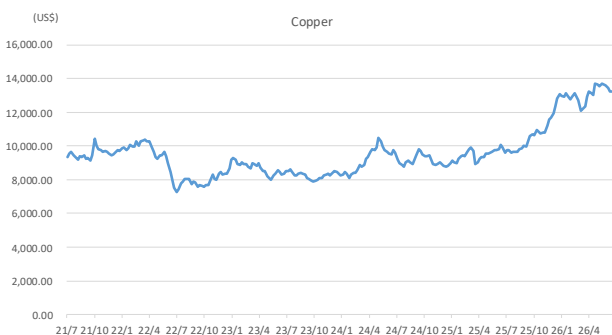
Platinum



Aluminum



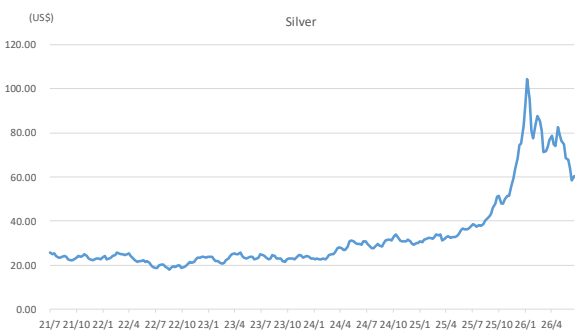
Copper



Gold



Silver



Note: Material prices as of 7/13/2025
Source: Bloomberg Finance L.P., J.P. Morgan.

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